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The Four Colour Theorem: An Introduction to Graph Theory



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Abstract

In this project we investigate the four colour theorem. The theorem asks, **can we colour a map in such a way that no neighbouring countries have the same colour using at most four colours?** We begin with the history of the theorem, seeing how it first sparked interest in mathematicians hundreds of years ago, and then move onto ideas of graph theory.

Building upon these discussions we will then examine the four colour theorem in more depth, covering the concepts required to reach its proof and review how it was first proved in 1976 via computer usage by mathematicians Kenneth Appel & Wolfgang Haken.

Contents

1	Introduction	4
2	History of the four colour theorem	6
3	Graph theory	18
3.1	Graph theory basics	19
3.1.1	Problems	22
3.2	Types of graphs	23
3.2.1	Problems	26
3.3	Graph alterations	27
3.3.1	Problems	32
3.4	Other concepts	35
3.4.1	Problems	39
4	The four colour theorem	44
4.1	Planar graphs	44
4.2	Colouring	54
4.2.1	Six colour theorem	57
4.2.2	Five colour theorem	57
4.2.3	Four colour theorem	59
5	Other surface embeddings	76
5.1	Polyhedra	76
5.2	Surfaces	81
6	Conclusion	87
7	Acknowledgements	89
8	Appendices	90
8.1	Research plan	90
8.1.1	The project	90

8.1.2	Action plan	91
8.1.3	Survey of the literature	94
8.2	Midway progress report	96
8.2.1	Introduction	96
8.2.2	What have I achieved so far?	96
8.2.3	What issues have arose or are anticipated?	99
8.2.4	Conclusion	100
8.3	Figures and notations	102
Bibliography		104

Chapter 1

Introduction

This project aims to leave the reader with an understanding of graph theory and how to apply its ideas to prove the four colour theorem. It begins with a consideration of the history behind the theorem, from when it was first mentioned on record, to when it was first proved some 200 years later. In order to grasp the ideas involved in this proof, we then cover aspects of graph theory.

The four colour theorem is thus; **a map can be coloured with no more than four colours in such a way that no two adjacent countries share a colour**. Graph theory allows us to view these maps with a different structure, one involving only edges and vertices. In turn, this allows us to identify properties and study our maps with increased ease. While both of these begin as simple concepts, we will see that it results in a complex proof.

On the surface one would be forgiven for not seeing the value in the four colour theorem. As we will see in Chapter 3, Leonhard Euler published a paper in 1736 which set out to solve the Königsberg bridge problem [29], this brought about the beginnings of graph theory. Although the four colour theorem was first noted 116 years later in 1852, it is considered to have vastly aided further development of graph theory [3, p.VII].

Real world advancements have resulted from this development of graph theory. For example, algorithms used in computer science are improving, and links to other areas of mathematics are beginning to surface and be useful [3, p.VIII]. Additionally, networking problems such as road networks, social media and even brain neuron connections can all be studied thanks to their shared properties with ideas from graph theory problems [34, p.3]. Indeed, the four colour theorem is not without its uses after all!

We have four main objectives in mind for this project. One is to explore the theorems' history, and another is cover some graph theory concepts including corresponding examples and problems. We also want to cover colouring theorems, again with corresponding examples, and lastly we want to explore how to apply colouring ideas to different surfaces. To achieve this we will predominantly follow the ideas and definitions used in the *Graph Theory* book by J. A. Bondy and U. S. R. Murty [3].

Chapter 2

History of the four colour theorem

One puzzle that confounded mathematicians for nearly a century is the idea ‘that if an area be partitioned in any manner into areas, these can be, with four colours only, coloured in such wise that in every case two attached areas have distinct colours’ [6], or more commonly known as, **the four colour theorem**.

Although some believe that this idea was originally observed by early cartographers, the first known instance of it was in 1852 by a 21 year old mathematician called Francis Guthrie. Francis went on to attempt a proof, and told his brother Frederick Guthrie about the four colour conjecture, as it then was known. Frederick is known to have said that Francis’ proof ‘did not seem altogether satisfactory’ [28], however, he still passed on the conjecture to Augustus De Morgan, a well-renowned mathematician whom Frederick had become a student of.

De Morgan then gave us the first written record detailing the four colour theorem, when he wrote to fellow mathematician William Hamilton on the 23rd October 1852. In the letter, he queried if Hamilton could provide any insight into what Frederick had shown him, as he was at a loss to explain it. Hamilton sent a short reply on October 26th, ‘I am not likely to attempt your quaternion of colour very soon [23].’

De Morgan kept spreading awareness of the problem and a few mathematicians, such as Charles Pierce, did attempt to prove it. Despite an interest in the conjecture continuing, there was not much progress noted until 1878, when Arthur Cayley queried at a meeting of the London Mathematical

Society if a proof has been found. The lack of significant progress reported spurred Cayley on to attempt it himself, and in 1879 he published *On the Colouring of Maps* [6].

In this paper, Cayley fails to find a proof but does succeed in highlighting the difficulty found in solving the four colour conjecture. The paper begins with Cayley saying that it is easy to see that four colours are necessary, for example, if we had a circle split into three sectors and then we enclosed it, the enclosed space would need to be a fourth colour, see Figure 2.1. Additionally, if we had a circle split into four sectors, we again could colour it with four colours but three would suffice. This showed Cayley, that in general, if the number of sectors is odd or even, then the number of colours required is four or three respectively.

Going forward we will refer to any sector/space etc. as a country, and the collection of all these countries as a map. Think of a map of the United States of America, each state is known as a country and all fifty of them make up the map.

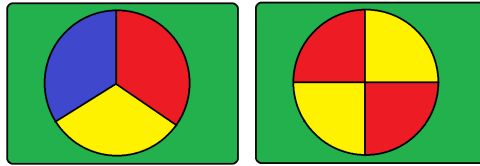


Figure 2.1: Cayley's first example - enclosed circles divided into three and four countries respectively.

Cayley's observation, however, does not provide a general proof, and gaining this is where the difficulty lies. To continue, he approaches the problem using mathematical induction. If we have n countries on a map, coloured as per the theorem with only four colours, and a $(n + 1)th$ country is added, we can not say that the new country can be successfully added without needing to alter the original colours first. For example, see Figure 2.2 where the enclosing white represents the $(n + 1)th$ country. It is not possible to colour this country that encloses the pre-existing map, as all four colours are present in its interior boundary, i.e. the new country is connected to all four colours already. Therefore, we would have to change the circle and colour it with only 2 or 3 colours to be able to colour the $(n + 1)th$ country.

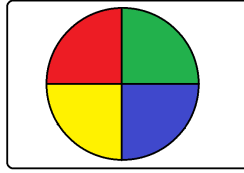


Figure 2.2: Cayley's second example - an enclosed circle that results in a map that cannot be four coloured, unless we make alterations first.

Alternatively, Cayley shows that instead of enclosing a map with an additional county, it is possible to insert the new $(n + 1)th$ country into the map and obtain the same result. This is achieved by inserting it at a point where it will be adjacent to all four colours. This leaves us with the same conundrum that is being highlighted, a boundary containing all four colours that needs changing to make the conjecture succeed. This way of visualising the difficulty is how we will now proceed.

Figure 2.3 shows us an example of this. We know the theorem states that it must be possible to colour the map, however we can see that it is not possible unless the colouring is altered. If we swap blue for red in one of the countries in the left hand side map, blue becomes available for the central country in the right. This works as we have changed the exterior boundary to only show three colours, though it could have been done with only two.

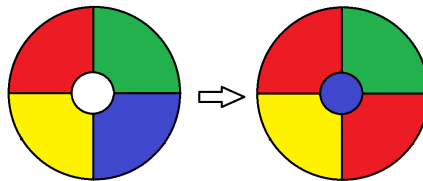


Figure 2.3: Cayley's third example - inserting a new country attached to each of the original countries.

These observations led Cayley to add a condition to the conjecture; no more than three colours can be present in the external boundary. This condition is what makes it difficult to prove. It would be insufficient to provide a general proof without also showing that it only works if it is possible to have the boundary condition in place too.

Despite this difficulty, Alfred Kempe on the 17th July 1879 announced that he had a proof. Kempe studied under Cayley, and it was per Cayley's suggestion

that he submitted his proof to the American Journal of Mathematics. Before we can fully explore this proof, however, we need to introduce some graph theory concepts. These concepts will be expanded upon in more detail in Chapter 3.

In graph theory, graphs are represented using **edges**, **vertices** and **faces**. Using our current analogy this translates to a graph being our map, the vertices (which can be coloured) being our countries, and the edges representing which countries are adjacent to one another. A face is an area that has been partitioned off by these edges and vertices. To help visualise this, Figure 2.4 shows the second circle from Figure 2.3 represented this way. For example, we can see the blue vertex has edges connecting to each of the others, indicating that it is adjacent to them all, as it indeed is.

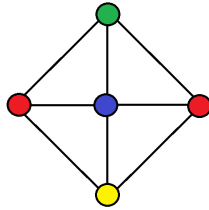


Figure 2.4: Representation of a map in graph theory.

In addition to this, we require the definition of a **planar graph**. This is a graph that can be drawn in the above way, without any lines crossing over. If the maximum number of edges are used in such a graph, meaning to add any more would make it nonplanar, then it is called **maximal planar**.

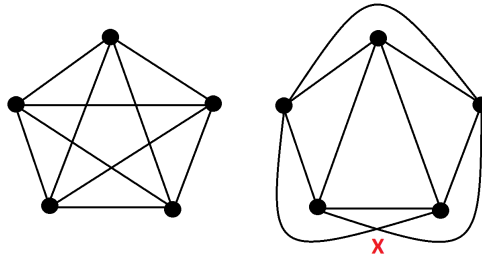


Figure 2.5: This is known as the pentatope graph. We can see that if we try drawing it without edges crossing that we fail, and therefore this is not a planar graph.

These definitions are important as Kempe recognised that the type of graph needed for the four colour conjecture is planar, and he combined this with **Euler's Formula** to prove that a planar graph 'must contain at least one country that is adjacent to five or fewer other countries [28].'

On a side note, Leonhard Euler was a prominent mathematician, who in 1736 wrote the first paper using graph theory [3, p.86], and we will be seeing more ideas from his work.

The number of edges connecting to a vertex is that vertex's **degree**, and here represents how many countries it is adjacent to. So Kempe's above idea in other words is, if a graph has all vertices 'degree 6 or higher, then the graph would not be planar [5].' This means that for Kempe's proof, he only had to show that it is possible to colour maps for cases that contain countries that are adjacent to five or fewer other countries.

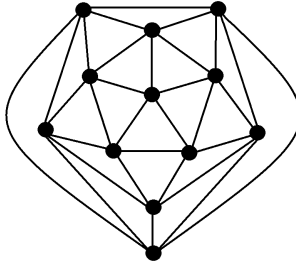


Figure 2.6: An example of a maximal planar graph, with every vertex having degree 5. It is not possible to draw a planar graph using degree 6 for every vertex.

The proof for this lemma, as phrased by Kempe in his original report, is as follows below.

Lemma 1. *Every map drawn on a simply connected surface must have a district with less than six boundaries [20].*

Proof. Euler's formula, mentioned above, will be proven later on. For now, take as fact that Euler's formula is $v - e + f = 2$, where v is the total number of vertices, e is the total number of edges, and f is the total number of faces.

We begin by considering maximal planar graphs, as we deduce that if they are colourable, then a planar graph with fewer edges will be too. Maximal planar graphs have the property that each face is bounded by 3 edges [3, p. 254], and that each edge is a boundary for 2 faces. This means that if we multiplied the number of faces by 3 to find the total number of edges, we would have counted each edge twice. So we have that $3f = 2e$ or,

$$f = \frac{2e}{3}$$

and substituting this into Euler's formula gives, $v - e + \frac{2e}{3} = 2$. Or equivalently,

$$e = 3v - 6.$$

As we began with maximal planar graphs, we know that we can remove edges and the graphs will remain planar. Therefore, for a planar graph we have,

$$e \leq 3v - 6.$$

We will now introduce a theorem, that again came from Euler and will be proven later.

Theorem 2. *The sum of the degrees of the vertices equals twice the number of edges, $\sum \deg(v) = 2e$.*

Returning to the lemma at hand, if we assume that all vertices have degree 6 then, by Theorem 2, we have $2e = 6v$ and by the same idea as before, this translates to $2e \leq 6v$ for planar graphs. We now have that $3v \leq e$, and we already know that $e \leq 3v - 6$ for planar graphs. Together, this implies that $0 \leq -6$, and so we have a contradiction, and a graph with degree 6 on each vertex cannot be planar. Therefore, we require that planar graphs have at least one vertex with degree less than six.

□

If we worked through Theorem 2 for the case of every vertex having degree 5, you would find that $v \geq 12$, so it would be possible to build a planar graph if every vertex had minimum degree 5. Looking back at Figure 2.6 we can see it has the minimum $v = 12$ required to be possible. For degree 4 you would get $v \geq 6$, degree 3 would be $v \geq 4$ and degree 2 is $v \geq 3$.

Kempe's proof was presented in an article with few visual examples and limited mathematical notation, so we will consider [28] which provides a more modern insight to Kempe's method. The proof we have just derived means that proof for the four colour conjecture needs only be found for cases when five or less countries border another.

To start with we have the base step; a map with four or less countries can be coloured using no more than four colours. Now we assume that this holds true for all maps with n countries, and let M be a map with $n + 1$ countries. We know that M must contain at least one country that is adjacent to five or fewer countries. Letting this country be known as X , we then build the map $M - X$ with the countries arranged in such a way that if we extended their boundaries into the excluded X they would meet at a point. See Figure 2.7.

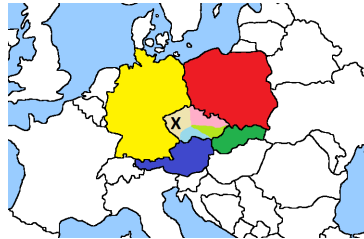


Figure 2.7: A map showing the Czech Republic as our X country, the lighter colours highlighting how the other countries could be extended to meet at one point.

Kempe's aim is to now reduce the number of colours used in the countries surrounding X , so that a colour is available for X . We can see for cases where X is surrounded by three or less countries that X can already be coloured. For cases with four or five surrounding countries, made up from three or less colours, again we can see X can already be satisfied. It remains that we need to consider cases where X is adjacent to four countries with four different colours, like in Figure 2.7, and cases where X is adjacent to exactly five countries coloured with four colours, like in Figure 2.8.



Figure 2.8: A map showing Slovakia as the X country, this time surrounded by five countries representing four colours.

To make his proof Kempe introduced a method called **Kempe chains**. A Kempe chain is a region made up from two continuously connected colours, for example in Figure 2.8 we have a red-yellow chain going from

Belgium-France-Italy-Austria. To state Kempe's argument simply, to make a map colourable you just need to interchange the colours in one or two chains that end in the boundary of X .

Case 1. X is adjacent to four countries.

Case 1 is split into two parts, the first being when non-adjacent regions in the boundary of X belong to different chains of the same colours.

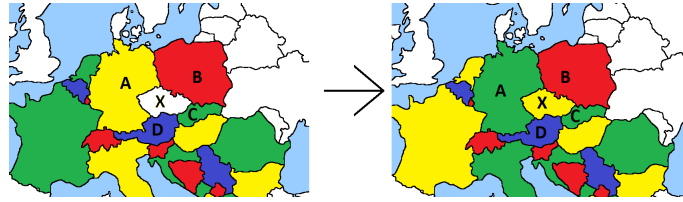


Figure 2.9: Case 1: Part 1. We have separate yellow-green chains coming from Germany A and Slovakia C , interchanging the colours in the Germany chain has meant X could be successfully coloured yellow.

Part 2 of Case 1 considers if the opposing sides belong to the same chain. In this instance, say we had a red-yellow chain, as the chain loops from one country to the other it would enclose countries that are in a blue-green chain preventing them from contacting the other blue-green chain that stems from the boundary. So to open up a colour for X , the colours in one of the blue-green chains would need inverting. See Figure 2.10.

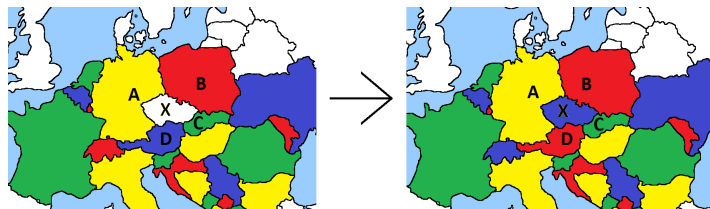


Figure 2.10: Case 1: Part 2. This time Germany A and Slovakia C share the same yellow-green chain, so interchanging their colours would not help. The chain has looped around Switzerland and Austria's D blue-red chain, thereby separating it from Poland's B chain. So we can invert a blue-red chain to get a successful colouring for X .

Case 2. X is adjacent to five countries.

Again, this case is split into two parts. Firstly, if we consider how the five countries are coloured, we can see that there will always be one country sandwiched between the same colour. Let the countries with the same colour be A and B , the colour in the middle be C , and the other two be D and E .

Then part 1 of this case is when, C belongs to a different, but same coloured, chain as D or E . In this instance, like in Case 1: Part 1, we interchange the colours in one of the chains. See Figure 2.11.

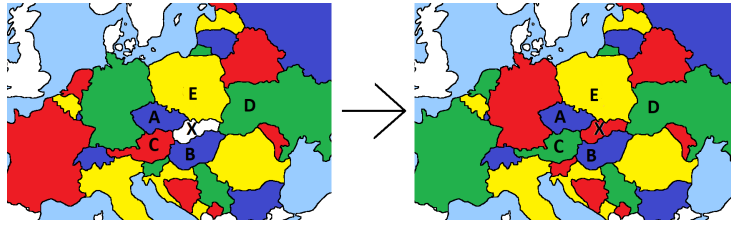


Figure 2.11: Case 2: Part 1. Here we have C being Austria, and its red-green chain is separate to the red-green chain of the Ukraine D . Therefore, interchanging one of these chains will free up a colour for X . In this example, C also has a separate red-yellow to Poland's red-yellow chain but that could have been connected, we just need one of D or E 's chains to not be connected to C 's.

The final part to Kempe's proof is the case when C belongs to the same chain as D and the same chain as E . In this scenario, the two chains keep countries A and B apart, and to free up a colour for X we need to invert both of their colours. See Figure 2.12.

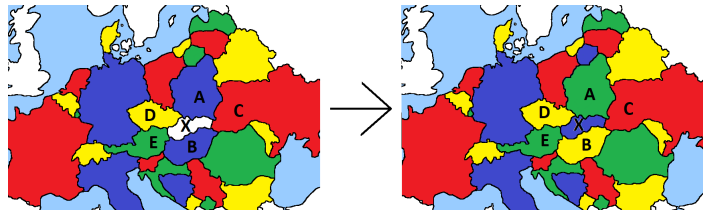


Figure 2.12: Case 2: Part 2. Here the Ukraine is the C country in the middle of A and B . It shares a red-yellow chain with D , the Czech Republic, and a red-green chain with E , Austria. This leaves us the green-blue chain of A and the blue-yellow chain of B segregated from one another. B , although it does not in this example have a yellow in the chain, is considered as blue-yellow as yellow is the colour it is not adjacent to from the boundary. Inverting A and B frees up the blue for X .

This concludes Kempe's proof from 1879, and for the eleven years following it, the four colour conjecture became the four colour theorem. Kempe received critical acclaim for his proof, and was elected as a fellow of the Royal Society in 1881, going on to be its treasurer and vice-president [23].

The error in Kempe's proof is not obvious. The fault is found in Case 2: Part 2, and it was Percy Heawood, a British mathematician, who produced a counterexample in 1890.

In Heawood's paper he states, 'The present article does not profess to give a proof of this original Theorem; in fact its aims are so far rather destructive' [17]. His proof was simply evidence that Kempe had failed to observe a third option within Case 2. Case 2 being that after finding chains from C to D and C to E , we invert the chains leading from A and B . However, Heawood found that it is possible that the chains from A and B might share a boundary, and this touching of the chains would mean that inverting them both would not be beneficial. At the point that they touch, if those two chains were inverted, we would have two adjacent countries that are both the original colour of A and B , and so not satisfying the four colour conjecture.

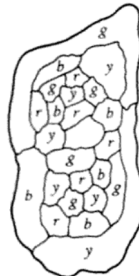


FIG. 6.19.

Figure 2.13: Heawood's original sketch which disproved Kempe's theorem [17].

Translating Figure 2.13 into the terms we have been considering, we have the X country in the center without a colour assigned to it. Then the C country is the blue one bordering X as it is sandwiched between two reds, representing A and B . Then we can see that C connects in a blue-yellow chain to the yellow D , and with a blue-green chain to the green E . This leaves, A (the red higher up) a red-green chain, and B and red-yellow chain. These two chains touch on the right hand side of the figure, where green and yellow are adjacent. If, as per Kempe's proof, we inverted both A and B 's chains, these touching countries would both become red, and so would not satisfy the theorem, and so it has been disproved.

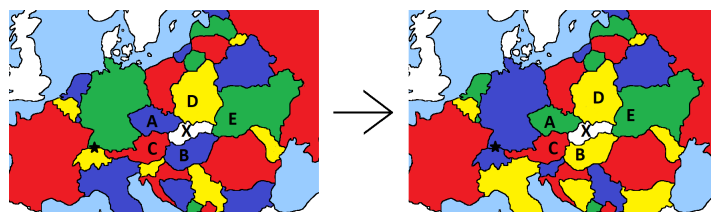


Figure 2.14: Example 2: The star denotes the point where the chains from A and B touch, and show how inverting them does not work for this case.

Heawood, unlike Kempe, did not receive any honour for his contribution, and in future discussions of Kempe's proof his findings were hard to find a record of, with some still believing that the proof stood strong [2]. Despite this, today Heawood is credited with the discovery as well as finding the first proof of the 5 colour theorem, which we will discover later. Kempe himself had reported the error to the London Mathematical Society, saying that he could not solve this 'case of failure [28].' Kempe's proof would remain the best attempt for almost a century.

In 1904 a new graph theory concept, **unavoidable sets**, was being widely studied. In 1922, Philip Franklin, an American mathematician, published work that used unavoidable sets, along with other ideas, to prove that any map M is four colourable with ≤ 25 countries. As more people got involved with the concept, the number increased. In 1926 it was proved for ≤ 27 countries, in 1940 for ≤ 35 countries, in 1970 for ≤ 39 and in early 1976 for ≤ 95 countries [23].

The next full proof came in 1976 from Kennet Appel & Wolfgang Haken. This proof renamed the four colour conjecture back into the four colour theorem for the last time, due to it remaining a successful and validated theorem.

This final proof expanded upon work started by Heinrich Heesch, a German mathematician. Heesch, during the 1960's and 70's, had begun to develop modern computer based proof methods. He used his own **method of discharging** for the four colour theorem, discussed in later chapters, and it was this that set vital groundwork for Appel & Haken to complete the proof [23]. Unfortunately for Heesch, he could not secure the supercomputer time needed to solve it himself.

Appel & Haken used the idea of unavoidable sets, the method of discharging, and **methods of reducibility** using Kempe chains, to check over 1400 cases on

a computer. This was the first theorem proof ever found via computer, as this made it difficult to verify the results by hand it was met with some scepticism. However, as other verification came about supporting Appel & Haken, the proof became more and more accepted.

In 1997, four mathematicians, Robertson, Sanders, Seymour & Thomas, published a report which improves on Appel & Haken’s algorithm so that less cases need to be considered. They clearly explain the concern few have about computer use in their conclusion, ‘our “proof” is not a proof in the traditional sense, because it contains steps that can never be verified by humans...These have to be taken on faith, and are conceivably a source of error [27].’

In 2005, Georges Gontier wrote his *Formal Proof—The Four-Color Theorem* report. In this, he outlines his use of the **Coq** programming language to formalise his version of the proof [14]. This lessens the worry about computer usage as the Coq proof assistance checks the proof process itself not just the mathematics being applied, meaning only the programming language has to be trusted to not create computational errors.

This concludes our history of the four colour theorem. We will now move on to explore graph theory in the next chapter, revisiting concepts not fully developed within this chapter. New ideas will be introduced to help develop an understanding of the mathematics needed to solve the final proof, in order to satisfy the objectives of this report.

Chapter 3

Graph theory

In Chapter 2, we have already introduced some graph theory concepts. This chapter aims to formalise those and introduce more, until we have laid out enough groundwork to discuss the four colour theorem in Chapter 4.

Graph theory was first seen in a paper written by Swiss mathematician Leonhard Euler, called the *Solutio problematis ad geometriam situs pertinentis*, which he wrote in Latin. Published in 1736, this fourteen page paper sets about solving the Königsberg bridge problem [29]. Königsberg at the time of Euler’s publication was part of Prussia, and today it is a small city called Kaliningrad in Russia. The city is split by a river that contains a loop, effectively splitting the city into four parts, and seven bridges were built to allow access to all districts.

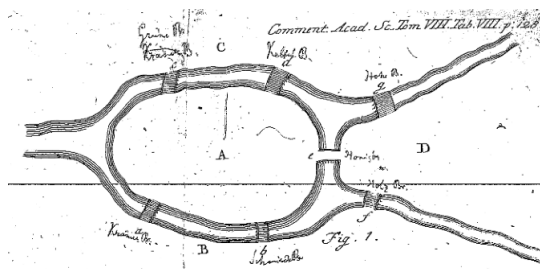


Figure 3.1: Euler’s original sketch, showing the locations of the seven bridges [11].

The problem was thus; is it possible to take a walk around the city, while only crossing each bridge once? Initially Euler was hesitant to attempt the problem, saying in a letter to the mayor of Danzig in Poland, ‘this type of solution bears little relationship to mathematics, and I do not understand why you expect a mathematician to produce it’ [24]. However, after considering the problem and concluding that it was not so trivial after all, Euler said, to an Italian mathematician called Giovanni Marinoni, it ‘seemed to me worthy of attention in that [neither] geometry, nor algebra, nor even the art of counting

was sufficient to solve it [24].’ Having to utilise new methods brought about the beginnings of graph theory.

We can see in Figure 3.1 that Euler labeled each distinct area A, B, C and D . Additionally, he said that the connection from A to B via the crossing of a bridge could be denoted (A, B) . This idea is the birth of **vertices** and **edges** in graph theory.

3.1 Graph theory basics

Firstly, we need to note that we are going to consider graphs of a finite nature, though infinite graphs do exist, as we will discover soon.

A graph G is the set of its vertices $v(G)$, and edges $e(G)$. The number we assign to these indicates the number of vertices and edges within graph G , and we can denote the entire graph as an ordered pair like this, $G = (v, e)$. Vertices are represented by a point when a graph is drawn, and the lines that connect them are the edges. If we can follow a root from a vertex to another, for all pairs of vertices, then the graph is **connected**. The following figure shows how the Königsberg map could be drawn using graph theory.

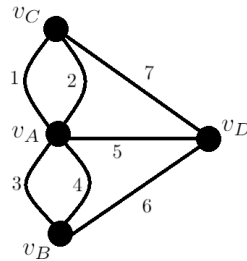


Figure 3.2: Königsberg map as a graph, $G = (4, 7)$. The subscript on the vertices indicates which area that vertex is from. To translate this back to the real map, we can see that there are two bridges connecting C to A , for example.

We would denote the set of vertices in Figure 3.2 as, $v(G) = \{v_A, v_B, v_C, v_D\}$, and the set of edges $e(G) = \{1, 2, 3, 4, 5, 6, 7\}$. When an edge links two vertices together then these vertices are said to be **adjacent**, the same applies for two edges that share a common vertex. For example, here we have v_A adjacent to v_D , and edges 1 and 7 being adjacent. The set of all vertices adjacent to a vertex is called its **neighbourhood**, and is denoted $N(v)$, vertex v_B has neighbourhood $N(v_B) = \{v_A, v_D\}$.

The **order** of a graph is the number of vertices that it contains, and similarly the **size** is the number of edges it has. The **degree** is the number of edges incident to a vertex, denoted $\deg(v)$ for an individual vertex, while the maximum degree found in a graph G is denoted $\Delta(G)$, and the minimum is $\delta(G)$. For Figure 3.2, we have $\deg(v_A) = \Delta(G) = 5$ and $\delta(G) = 3$ at all the others.

It is possible for an edge to **loop** back on itself to the same vertex, and even though it is the same edge, this would count as two when considering that vertex's degree value. We can see in the Figure 3.2 an example of **parallel edges**, these are edges sharing the same pair of vertices, edges 3 and 4 are parallel edges. A graph without loops or parallel edges is a **simple** graph, we will predominantly be considering simple graphs throughout this report.

Another definition that we already saw was the **face**, $f(G)$. A face exists when the edges of a graph enclose an area, the degree of this face is the number of edges it is bound by. If both sides of an edge are present in a face then it is counted twice when considering that faces degree. When faces are being counted, say we were using Euler's formula, we also consider the external area of the graph as a face. We call this the **infinite face**. Therefore, we can say for Figure 3.3 below, that the graph on the left has 10 faces, $f(G) = 10$.

Figure 3.3 also shows an example of an **infinite graph** on the right, we have already noted that these will not be fully considered here, as we are working with **finite graphs**. However it is useful to show them, as when thinking about graph theory many concepts can be applied to them in the same way, such as degree, and others with some slight alteration.

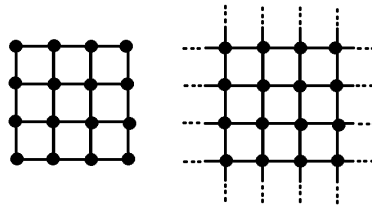


Figure 3.3: Finite graph on the left, infinite on the right. This particular infinite graph is known as the **square lattice** [3, p. 36]

We now have a couple of more complex ideas to introduce, **isomorphisms** and **automorphisms**. A graph G is **identical** to a graph H , if it has the same edge set, $e(G) = e(H)$, the same vertex set, $v(G) = v(H)$, and its edges connect the same vertices. To know which edges connect to which vertices we need the

incidence function, ψ_G . We use $\psi_G(e) = \{v_a, v_b\}$ to say that edge e joins vertex v_a and vertex v_b [3, p. 2]. So for identical graphs G and H we would have $\psi_G = \psi_H$.

The incidence function, ψ_G , of graph G in Figure 3.4 is, defined by

$$\psi_G(a) = \{1, 3\}, \quad \psi_G(b) = \{1, 5\}, \quad \psi_G(c) = \{1, 5\}, \quad \psi_G(d) = \{4, 5\},$$

$$\psi_G(e) = \{2, 4\}, \quad \psi_G(f) = \{2, 5\}, \quad \psi_G(i) = \{3, 3\}, \quad \psi_G(j) = \{3, 4\}.$$

We can see that $\psi_G(b)$ and $\psi_G(c)$ are incident to the same vertices and so these edges are parallel. Also, $\psi_G(i)$ is only incident to one vertex and so it is a loop.

Clearly, identical graphs could be visualised with identical diagrams. In the case where we have diagrams of graphs G and H that are the same, yet do not have matching edge or vertex sets, then these are not identical; and we have graphs that are said to be **isomorphic** to one another, denoted $G \cong H$.

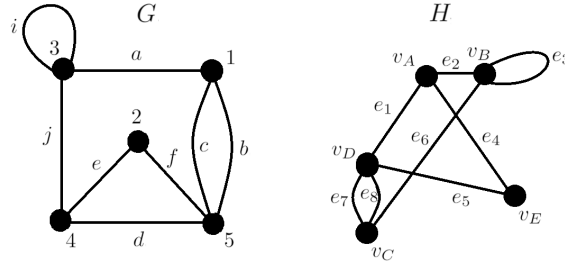


Figure 3.4: Graphs G and H have different notation for their vertex and edge sets so are not identical. Yet we can see that they have the same structure, as evident if we draw one of them differently, and so $G \cong H$.

More officially, we have an isomorphism $G \cong H$, if we have a bijection between G and H . A bijection means that we have a mapping from one set to another, that is injective: each element is connected to only one in the other set, and surjective: every element has a connection in the other set. A bijection is denoted $\theta : G \rightarrow H$. In graph terms we need the following bijections, $\theta : v(G) \rightarrow v(H)$ and $\phi : e(G) \rightarrow e(H)$; and that the adjacent vertices remain adjacent following the mapping, $\psi_G(e) = \{a, b\}$ if and only if $\psi_H(\theta(e)) = \{\theta(a), \theta(b)\}$ [3, p. 12].

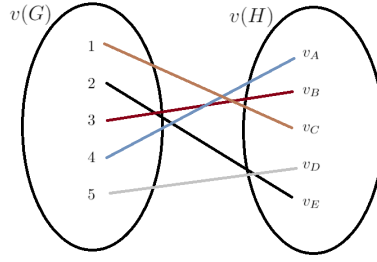


Figure 3.5: A visual representation of the $\theta : v(G) \rightarrow v(H)$ bijection between our graph G and graph H .

An automorphism of a graph is very similar to an isomorphism, except that instead of G mapping onto another set H , G maps onto itself. These isomorphisms of a graph onto itself give us the symmetries of that graph.

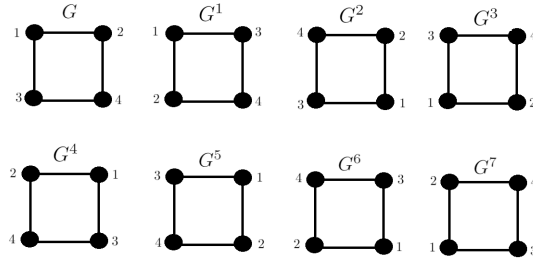


Figure 3.6: The eight automorphisms of a simple square graph, G itself is counted. Let us consider the graph G^4 , where G is the graph $\{1, 2, 3, 4\}$, G^4 is the isomorphism $\theta : G \rightarrow G$ that gives us $\{2, 1, 4, 3\}$. This represents the symmetry of reflecting in the y -axis.

$Aut(G)$ denotes the set of automorphisms and $aut(G)$ the quantity of them. For example, the graph G in Figure 3.6 has $aut(G) = 8$ and $Aut(G) = \{(1, 2, 3, 4), (1, 3, 2, 4), (4, 2, 3, 1), (3, 4, 1, 2), (2, 1, 4, 3), (3, 1, 4, 2), (4, 3, 2, 1), (4, 3, 2, 1), (2, 4, 1, 3)\}$.

3.1.1 Problems

Our first problem section revisits a theorem we saw back in Chapter 2 and covers it more formally.

Theorem 2. *For a graph G with e being the total number of edges, the sum of the degrees of the vertices equals twice the number of edges, $\sum \deg(v) = 2e$.*

Proof. Using proof by induction [32], we first consider the case of $e = 0$. Then we have $\sum \deg(v) = 0$ and so the theorem holds true.

Now suppose that we have proved the case for a graph with $e - 1$ edges. Let us consider a graph G with e edges, and denote one of them e_1 . If we remove e_1 we create the graph $G \setminus e_1$, then this has $e - 1$ edges and so we can assume the theorem holds for it.

Graph $G \setminus e_1$ then has a total sum of degrees equal to $2(e - 1) = 2e - 2$. So our graph G with e_1 added back into it, has e edges and increases the total sum of degrees by 2 giving us $\sum \deg(v) = (2e - 2) + 2 = 2e$, proving the theorem.

□

Corollary 2.1 *It can be observed that Theorem 2 tells us that the sum of degrees is always an even number.*

Corollary 2.2 *It can be observed that Theorem 2 tells us that the quantity of vertices with an odd number of degrees is even.*

The corollaries tell us that a graph with 4 vertices of degree 2, 2, 3, and 4 cannot exist, while a graph with 5 vertices of degree 3, 5, 5, 6, and 7 could.

3.2 Types of graphs

Graphs come in a variety of types, this can depend on the number of edges and vertices, what structure they form, how they relate to one another...etc. In this section, we will illustrate a handful of graph types with complementing examples.

We have already seen that a graph without parallel edges or loops is a **simple graph**. In the case that any pair of vertices in a simple graph are adjacent to one another, we have a **complete graph**. Using n to represent the number of vertices, i.e. the order of the graph, we denote complete graphs as K_n . In contrast, if no vertices are connected and therefore none are adjacent, then we have an empty edge set $e(G) = \emptyset$. This is called an **empty graph**, denoted N_n .

When we have a vertex without a connecting edge, such as all the vertices in an empty graph, then it has $\deg(v) = 0$, and is known as an **isolated vertex**.

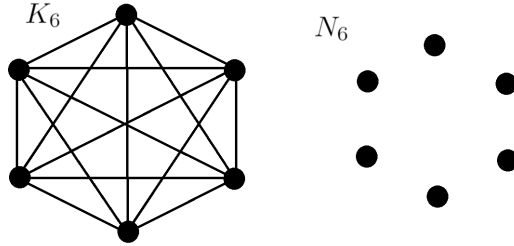


Figure 3.7: Examples of a complete graph and an empty graph, both with six vertices, $n = 6$.

Complete graphs have the property that their size is,

$$e(K_n) = \frac{n(n-1)}{2}$$

which we can intuitively understand. As every vertex is adjacent to all others then they must have degree $n - 1$, and we have n of them. Giving $n(n - 1)$ edges, all of which are counted twice as they are incident to two vertices, and so we divide by two to get the total number of edges. For our K_6 , we can see that we then have $e = \frac{6 \times 5}{2} = 15$.

When a graph has the same degree for all vertices then it is a **regular graph**. It is commonly denoted as k -regular, where k is the degree value. We just saw that all vertices in a complete graph have the same $n - 1$ degree, and so complete graphs are $(n - 1)$ -regular. A **cubic graph** is a graph that is 3-regular [3, p.7].

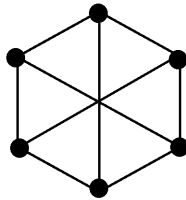


Figure 3.8: Example of a cubic graph with six vertices, and nine edges.

Now we have three types of graphs that get their name from the shape that they form, the **circular graph**, the **star graph**, and the **wheel graph**.

A circular graph C_n , is one where every vertex has degree two, and they link in such a way that that form a single closed chain, with higher and higher values of n this appears like a circle, hence the name. For example, if we had vertex set $v(C_n) = \{1, 2, 3, \dots, n-1, n\}$, then the edges would be adjacent as the follow set determines: $e(C_n) = \{\{1, 2\}, \{2, 3\}, \dots, \{n-1, n\}\}$.

A star graph S_n , is one where we have a single vertex with degree $n-1$, and all the others adjacent to it have degree one. A vertex with degree one is called a **pendant vertex** and its connecting edge is a **pendant edge** [15, p. 98]. A wheel graph W_n , has a vertex known as the hub in its center. The hub has degree $n-1$, and so connects to all other vertices. The other vertices, then form a chain, like the circular graph, giving the appearance of a wheel.

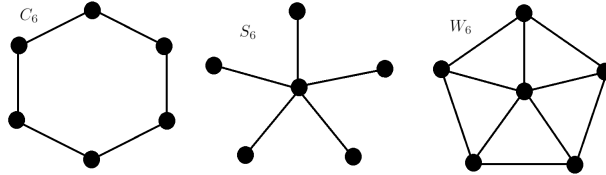


Figure 3.9: Examples of a circular graph, a star graph, and a wheel graph. All with six vertices, $n = 6$.

Our final graph type that is distinguished because of its structure, is the **lattice graph**. We have already seen an example of these back in Figure 3.3, showing the square lattice in a finite form and an infinite one. These graphs could be made from any shape that allows a regular tiling pattern [30].

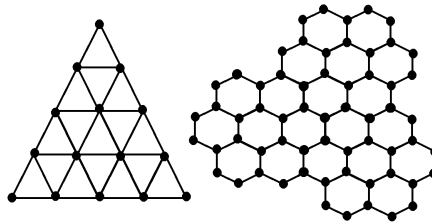


Figure 3.10: A finite triangular lattice and hexagonal lattice example.

We will now end this section with **bipartite graphs**. A graph is bipartite if we can separate its vertex set into two subsets, X with vertices $\{x_1, x_2, \dots, x_m\}$ and Y with vertices $\{y_1, y_2, \dots, y_n\}$, where edges can only join vertices of subset X and subset Y together. No vertices within the same set are adjacent [15, p.78]. X and Y are known as the **parts** of the **partition** (X, Y) of graph G [3,

p.4]. A bipartite graph is then denoted $G[X, Y]$. If we look at the star graph S_6 above in Figure 3.9, we can see that it is the bipartite graph $G[1, 5]$.

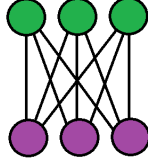


Figure 3.11: In this instance we have a green part X and a purple part Y . We can see that no vertices within X or Y are adjacent to vertices in their corresponding parts, and so this is a bipartite graph. Additionally, as every vertex is adjacent to every vertex in the opposing set, we can say that this is a complete bipartite graph. This also means each vertex has degree three in this case, as the subsets have order three, and so it is 3-regular too. Following the complete graph notation we could then label complete bipartite graphs as $K_{m,n}$, where m and n are the number of vertices in each subset, and this particular graph is $K_{3,3}$.

3.2.1 Problems

The three utilities problem can be found described in a variety of ways, but most commonly it is given as: Can you install gas, water and electricity to houses A, B, and C without any of the pipes crossing over?

This problem appears in this subsection, as we can see that it can be illustrated by the complete bipartite graph $K_{3,3}$. Thinking back to the definition of a planar graph in Chapter 2, this problem can be solved by determining if $K_{3,3}$ is a planar graph, as then no edges (representing the pipes) can cross over.

Following a method for proof from [21] we find that the three utilities problem does not have a solution.

Proof. Let e be the total number of edges in a planar graph. For planar graphs we previously saw that each edge is bounded by two faces. So if we have faces that are bounded by n edges, then we have $nf = 2e$. Using the $f = \frac{2e}{n}$ given by this, and substituting it into Euler's formula $v - e + f = 2$ gives us:

$$e = \frac{n(v - 2)}{(n - 2)}.$$

Now, it is hard to tell how many edges the faces are bound by in $K_{3,3}$ until it is drawn as if it was in three dimensions, see Figure 3.12 below.

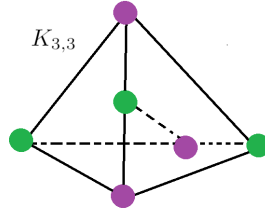


Figure 3.12: When drawn in three dimensions, we can see that the faces are bound by $n = 4$ edges.

Inputting this into our formula gives, $e = \frac{4(6-2)}{(4-2)} = 8$. However, when we actually count the edges in $K_{3,3}$ we find that there are 9 of them. Therefore, we have a contradiction and it cannot be a planar graph, and so the three utilities problem does not have a solution.

□

3.3 Graph alterations

In this section we will look at concepts that arise when something alters or provides a different version of a graph G .

We will begin with **subgraphs** and **supergraphs**. If we have a graph G , with $v(G)$ and $e(G)$, and a graph H with $v(H)$ and $e(H)$, then H is a subgraph of G if its edge and vertex sets are both contained in the vertex and edge sets of G , and the connections in the incidence function are maintained; $v(H) \subseteq v(G)$, $e(H) \subseteq e(G)$. A graph that contains the subgraph is the supergraph.

Subgraphs are denoted by $H \subseteq G$, meaning H is a subgraph contained in G , and supergraphs are denoted $G \supseteq H$, meaning G is the supergraph containing H . A graph G is a subgraph and a supergraph of itself. When we have $H \subseteq G$ but $H \neq G$, i.e. $e(H) \subsetneq e(G)$ or $v(H) \subsetneq v(G)$, then H is a **proper subgraph** of G .

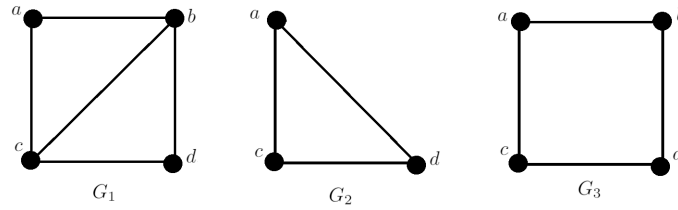


Figure 3.13: Here we can see that G_2 has $v(G_2) = \{a, c, d\}$ which is contained in the vertex set of G_1 , $v(G_1) = \{a, b, c, d\}$. However, the edge that connects a and d , is not in G_1 , and so its edge set is not contained in G_1 . Therefore, $G_2 \not\subseteq G_1$. For graph G_3 we can see that both of its edge and vertex sets are contained in the edge and vertex sets of G_1 , and so $G_3 \subseteq G_1$ and $G_1 \supseteq G_3$.

Subgraphs can be formed from supergraphs, by either **vertex deletion**, or **edge deletion** [3, p.40]. In the next figure, we see examples of this.

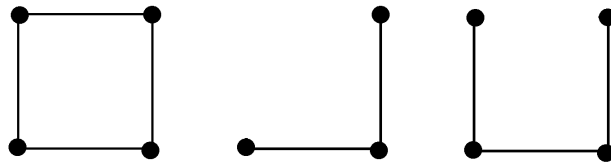


Figure 3.14: Let the left hand graph be G , then the central graph is a subgraph of G created following a single vertex deletion, the graph $G - v$. In the right hand graph we have a subgraph formed with a single edge deletion, $G \setminus e$. These two are known as a **vertex-deleted subgraph** and an **edge-deleted subgraph**, respectively. Any subgraph of G can be found by repeating a series of these deletions.

In the case where every edge which is incident to the vertices of the subgraph has not been deleted, we have an **induced subgraph**. In Figure 3.14 above, the central subgraph has lost no edges that connect the remaining vertices within the vertex subset, and so it is induced. However, as the supergraph of the right hand one had an edge connecting the top two vertices and the subgraph does not, then it is not induced [15, p.88].

Next we have the ideas of **union**, **intersection** and **ring** [26, p.12]. There is a union between two graphs G and H , when their edge and vertex sets are combined into one graph. Union is denoted with a $\cup$, so for a union $G \cup H$ we are after $v(G) \cup v(H)$ and $e(G) \cup e(H)$.

The intersection of graphs G and H , denoted $G \cap H$, is the graph showing all the shared edges and vertices from the two graphs, found from the intersections of their vertex and edge sets $v(G) \cap v(H)$ and $e(G) \cap e(H)$.

A ring, uses the union of vertices $G \cup H$, but then only keeps the edges that are in G or the edges in H . No edges in a ring are found in both G and H . A ring is denoted $G \oplus H$.

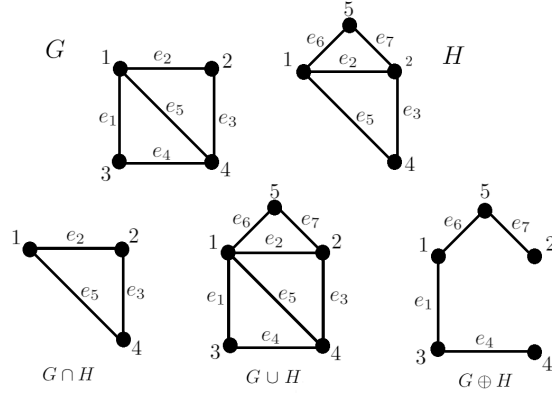


Figure 3.15: Here we can see graph G and H alongside their intersection $G \cap H$, union $G \cup H$ and ring $G \oplus H$.

A graph G that is split into two subgraphs G_1 and G_2 , such that $G_1 \cup G_2 = G$, and $G_1 \cap G_2$ is an empty graph, has been **decomposed** into two the subgraphs.

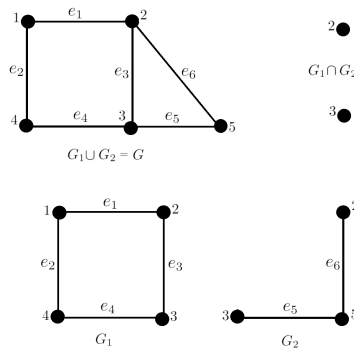


Figure 3.16: We can see that $G_1 \cup G_2 = G$, and that if we intersect G_1 and G_2 that they have no shared edges, and so $G_1 \cap G_2$ produces an empty graph. Therefore, we can say that G is decomposed by the two subgraphs.

We have already seen that a graph where every vertex can be linked to another via edges is a connected graph. When we have two connected graphs

that share no vertices, and so are not connected to each other, then they are **disjoint**. If we have two graphs G and H that are disjoint, their **disjoint union** is denoted $G + H$ instead of $G \cup H$ [3, p.29]. Disjoint graphs that form G after union, are called the **components** of G .

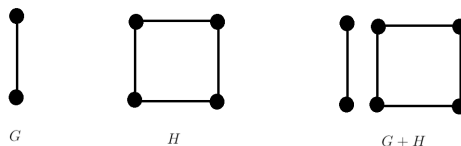


Figure 3.17: G and H are disjoint and are the components of the graph $G + H$.

It is also possible for two graphs to not share any edges, but share the same vertex set, and so be **edge-disjoint**.

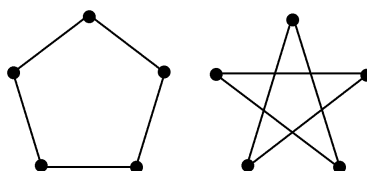


Figure 3.18: These two graphs are edge-disjoint.

Edge-disjoint graphs are similar in concept to a graphs **complement** (also known as a graphs inverse). A graphs complement, denoted $\overline{G}$, is a graph with the same vertex set as G but with all the opposite edges to G . Vertices that were not adjacent in G are in $\overline{G}$, and vertices that were adjacent in G are not in $\overline{G}$. An edge-disjoint graph does not have to contain all the ‘missing’ edges from G , just one would be sufficient. In this case though, the edge-disjoint example in Figure 3.18 does also show a graph with its complement. We can consider the empty graph N_n to be the complement of the complete graph K_n .

If a graph G contains an edge e that when removed would turn G into a disconnected graph, i.e. $G \setminus e$ has one more component than G , then this edge e is called a **cut edge** [3, p.85]. Looking back at G_2 in Figure 3.16, we can see that e_5 is a cut edge of G_2 . We also can have a **cut vertex** such that $G - v$ results in more components than G had, consider again Figure 3.16 to see that 5 is a cut vertex of G_2 .

The minimum number of vertices k that need removing to increase the number of components within a graph G is that graph’s **vertex connectivity**,

denoted $\kappa(G)$, and we say that for $\kappa(G) = k$ the graph is k -**connected**. Similarly, we have the minimum number of edge deletions required to increase the number of components in G , denoted $\lambda(G)$, which gives us the **edge connectivity**. A graph is k -**edge connected** for $\lambda(G) = k$ [26, p.119].

To have a 1-connected graph is to have a connected graph that breaks into more components when 1 vertex is removed, for example tree graphs, seen later on, are all 1-connected.

Furthermore, we have **blocks**. A **block** is a connected subgraph H of a graph G that does not contain a cut vertex. To count as a block, H must be maximal. This means that adding any vertex would mean that there was a cut vertex in H , so H is the largest it can be without containing a cut vertex. Any graph G can be split into its blocks, as seen in the next figure.

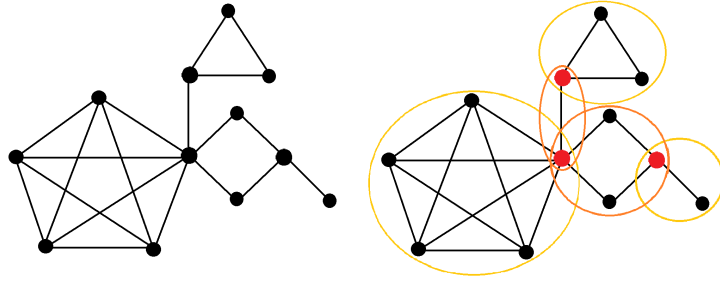


Figure 3.19: Here we see a graph G on the left. On the right G is repeated with its blocks circled in yellow or orange, and its cut vertices highlighted in red. We can see K_5 , C_3 and C_4 form blocks in this graph.

The last few concepts we will see in this graph alterations subsection are, **identify**, **contract** and **split**.

If vertex v_1 is not adjacent to v_2 in graph G , but the graph was drawn using a single vertex to represent v_1 and v_2 whilst maintaining all their edges, then we have **identified** those two vertices. The graph which includes identified vertices is denoted $G \setminus \{v_1, v_2\}$.

If we perform an edge deletion, and the vertices that were adjacent to this edge are then identified, then we have **contracted** that edge. Performing the inverse of this **splits** a vertex, turning it back into two vertices and adding an edge to make them adjacent to one another in the graph.

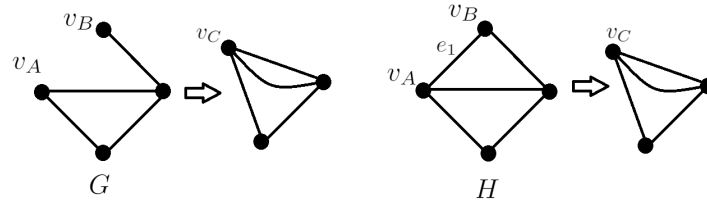


Figure 3.20: Here we have that vertices v_A and v_B have been identified in graph G to form v_C , and edge e_1 has been contracted in graph H . If we were to split v_C in the right hand side graph, we would not necessarily get H returned. We could split it so that we have some parallel edges from v_B instead of the edge leading from v_A that H has, and so splittings are not unique.

3.3.1 Problems

In this problems section we will be looking at another well known graph theory problem, Hall's marriage theorem. Named so as it is commonly described using marriage, the theorem asks, is it possible to find a desirable marriage for every man in a group of men, given that each has a preferred selection of choices from a group of women?

The theorem requires some further definitions. Firstly, a **matching** M in a graph G . This M is a set of edges that share no vertices, and not all vertices need to be included in M [3, p.412]. The vertices that an edge in M join are said to be **matched** under M . If every vertex of G is included in M then we have a **perfect matching**, and a **maximum matching** is one that uses as many vertices as possible [3, p.414].

The theorem was solved using this idea of matching but applied to a bipartite graph $G[X, Y]$, where men represent set X and women set Y . A matching in a bipartite graph would be the set of edges that match vertices in X and Y together. A **complete matching** in a bipartite graph is one where every vertex from one set is in M .

So Hall's Theorem in graph theory terms asks, can you find a complete matching in a bipartite graph $G[X, Y]$ that includes every vertex of X , given that each X can only match to a particular subset of Y ?

For example, let $X = \{a, b, c\}$ and $Y = \{1, 2, 3\}$. Also, the subsets that the vertices of X wish to join to in Y are $a \rightarrow \{1, 3\}$, $b \rightarrow \{2\}$, and $c \rightarrow \{1, 2\}$. As b can only match to 2, that leaves c matched to 1 and a matched to 3, giving a perfect matching $M = \{\{a3\}, \{b2\}, \{c1\}\}$.

Theorem 3. *Hall's Theorem.* *A bipartite graph $G[X, Y]$ has a complete matching satisfying every vertex in X if and only if for every subset of X , $x \subseteq X$, the subset of vertices in Y they can match to, $y(x) \subseteq Y$, has at least the same number of vertices or more:*

$$|y(x)| \geq |x| \quad \text{for all } x \subseteq X$$

Proof. It is clear to see that Hall's Theorem is required to find a complete matching, if x tries to match to a smaller sized $y(x)$ then there will not be enough vertices to match to.

Let us consider a graph $G[X, Y]$ such that $|X| = |Y|$ and assume that the marriage condition holds so that a complete matching can be found. Then we will apply proof by induction. For subset $x = 1$ we can have a complete matching with any subset $y(x)$, and so the base step holds.

Now if we assume it holds true for $x + 1$ and let u denote the vertex that has been added. Following the method from [18], we now have three cases to consider.

Case 1. Vertex u can happily match to a vertex outside of the $y(x)$ subset that x can match to. Therefore, a matching for $x + 1$ can clearly be found.

Case 2. Vertex u can only match to vertices in $y(x)$, and $|y(x)| = |x|$ so there are no vertices left for it to match to. In this case it is impossible to find a matching for $x + 1$. Agreeing with the theorem that no matching can be found if $|y(x)| < |x|$.

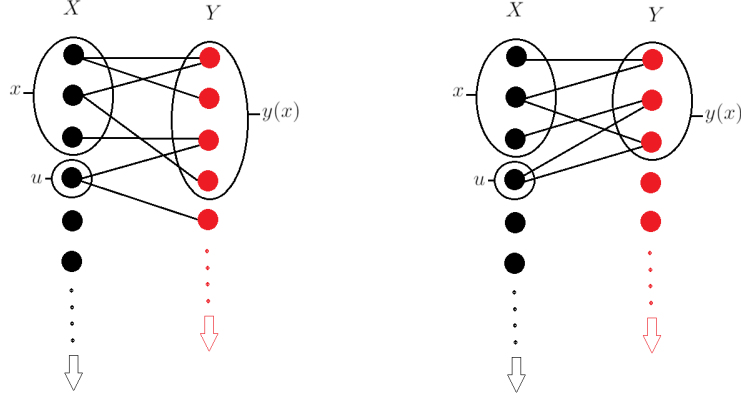


Figure 3.21: On the left hand side we see an example of Case 1, and on the right we have Case 2.

Case 3. Vertex u can only match to vertices in $y(x)$ and $|y(x)| > |x|$. As we have assumed that we have a complete matching, it will be possible to match every $|x|$ sized subset in G . Then if we take an $|x| - 1$ subset and add u , it is the same size as $|x|$ and by our assumption will have a complete matching.

To show that this means an $|x| + 1$ matching will succeed, let us consider this subset with one removed and u added, and call it x' . Let the subset of $y(x)$ that is used for the matching of x' be $y(x)'$, so that $|x'| = |y(x)'|$. Finally, let the vertex we removed from x to make room for u be vertex v , see Figure 3.22.

If v is connected to a vertex in $y(x)$ outside of $y(x)'$ then, as in Case 1, there are no issues matching it (the green matching in Figure 3.22).

If the only vertices v can match to are found in $y(x)'$ then, as $|x \cup u| > |y(x)'|$, and we have assumed that a matching does exist in G for this, there must exist a vertex in x' such that this vertex can find a matching outside of $y(x)'$ (the blue matching in Figure 3.22). This would mean that we could add v into x' and remove the vertex that can make the blue match outside of $y(x)'$ and still have a complete matching. Then as the vertex removed has a matching outside of $y(x)'$ it could be added to x' , creating $|x' + 1| = |x + 1|$ with a complete matching and so the theorem holds for any graph G .

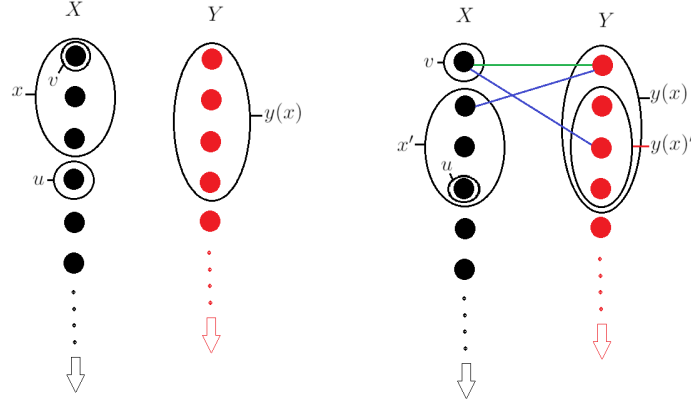


Figure 3.22: On the left we have graph G , and on the right how we can successfully add v .

□

3.4 Other concepts

In this final section of the chapter we will investigate some further ideas from graph theory, these will be useful in solving the Königsberg problem later on.

To find this proof Euler introduced some new ideas. Firstly, we have the concept of a **walk**, this is when we traverse a graph using a sequence of edges and vertices. A walk can contain repeated edges and vertices. A walk would be denoted as $W = \{v_0 e_1 v_1 \dots e_{l-1} v_{l-1} e_l v_l\}$. The starting point v_0 is the **initial vertex**, and v_l is the **terminal vertex**, every vertex in between is an **internal vertex** [3, p.80]. The **length** of the walk is quantity of vertices visited during the walk, excluding the initial vertex.

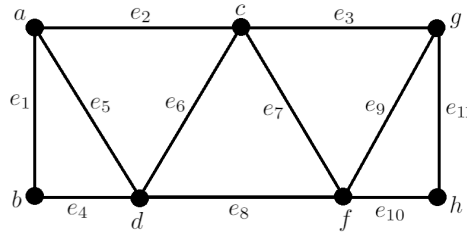


Figure 3.23: For example, we could take a walk $W = \{ae_5de_8fe_{10}he_{11}g\}$. As a is our initial vertex, and g is the terminal, we could call this an ag – walk. If we want to take a **segment** of this W , say the segment d to h , we would call it a dh – walk and denote it $dWh = \{de_8fe_{10}h\}$ [3, p.80].

Expanding these ideas, we have **trails** and **tours**. A trail is when we walk around a graph without repeating an edge. A **Euler trail**, is a trail around a graph where every edge is included only once [3, p.86].

When we have a walk that begins and ends at the same vertex, then this route is called **closed**. A **tour** is a closed walk that uses every edge at least once. If a tour includes every edge at most once, then it is a **Euler tour**. A graph with a Euler tour is **eulerian** [3, p.86].

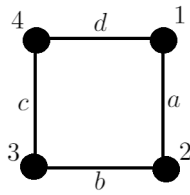


Figure 3.24: A simple example to illustrate trails and tours. For this graph, let's say we walk around it in the following order of vertices and edges: $\{1, a, 2, b, 3, c, 4\}$. Then, we have repeated no edges, and do not start and end at the same vertex, and so we have a trail. If we walked for longer and did this: $\{1, a, 2, b, 3, c, 4, d, 1\}$, then we have used every edge exactly once, and started and ended at the same vertex, and so have a Euler tour. Therefore, this is an eulerian graph.

Furthermore, a **path** is a walk without any repeated vertices except when it is **closed**. When closed, the initial vertex is the same as the terminal vertex, and the path forms a **cycle**. A path of length k , is a k -path or a k -cycle if closed. If every vertex was included in a path then we have a **Hamiltonian path** and a closed Hamiltonian path is a **Hamiltonian cycle** [3, p.472].

Moving on to a different concept, we now introduce the idea of **weight**. Weights are assigned to edges to indicate the 'cost' of that edge. The higher the weight, the more costly it is to walk along that edge, the weight could indicate the length of a road or how expensive that route is, for example. A **weighted graph** is denoted (G, w) with w being the total weight, and the specific weight assigned to each edge is denoted $w(e)$. We will see how weights can be useful in the first problem later on.

Next, we will discuss the practicality of studying graphs. It is clear to see that as we consider increasingly larger and larger graphs, that it becomes more challenging to illustrate them. Graphs can instead be described in matrix form with **adjacency matrices** and **incidence matrices**. This representation

creates the graph information in a more digestible way that can be worked with in computers, allowing more complex graphs to be studied.

Let us consider both types of matrix for the Königsberg bridge graph. An extra edge has been added to illustrate how loops are included in the matrices.

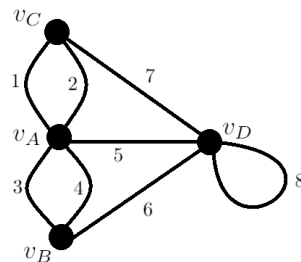


Figure 3.25: Reminder of the Königsberg map as a graph, with 8 representing a new loop edge.

The incidence matrix of a graph G , is an $n \times m$ matrix, where n is the graph order, and m is the graph size. A 0 entry in the matrix would mean that this pair of edge and vertex are not incident to one another, 1 would mean that they are incident once, and 2 would mean that they are incident twice, implying that the edge was a loop onto that vertex.

The adjacency matrix is an $n \times n$ matrix, and shows how many times vertices are adjacent to one another. The values indicate how many edges connect one vertex to another, 2 could mean parallel edges or a loop. Below shows both matrices for the graph in Figure 3.25.

	1	2	3	4	5	6	7	8
v_A	1	1	1	1	1	0	0	0
v_B	0	0	1	1	0	1	0	0
v_C	1	1	0	0	0	0	1	0
v_D	0	0	0	0	1	1	1	2

Incidence matrix

	v_A	v_B	v_C	v_D
v_A	0	2	2	1
v_B	2	0	0	1
v_C	2	0	0	1
v_D	1	1	1	2

Adjacency matrix

We can see that the adjacency matrix is the more compact of the two, and we can go one step further with an **adjacency list** which simply is a list of the vertices that one is adjacent to, for example $v_A = \{v_B v_C v_D\}$. This is how most graphs are stored in computers [3, p.6].

The final concept we will discuss as it leads us to the proof of Euler's formula we saw earlier on in Chapter 2. This is the idea of **trees**.

We have just seen in this section what a cycle is. When a graph contains no cycles that graph is known as an **acyclic graph**. If we have a simple connected acyclic graph then we call it a **tree**. Trees have the property that they only have the infinite face (as having a cycle would add a face), and they must contain vertices with $\deg(v) = 1$, known as **leaves** [3, p.99]. Having a disjoint acyclic graph gives us a graph known as a **forest** with trees as its components.

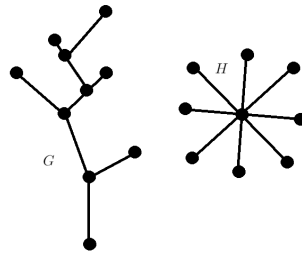


Figure 3.26: We have two trees, G and H . We can see that H is the star graph S_9 . The graph $G + H$ would be a forest.

Theorem 4. Euler's Formula. *For a connected planar graph G we have the relation*

$$v(G) - e(G) + f(G) = 2,$$

where $v(G)$, $e(G)$ and $f(G)$ are the quantity of vertices, edges and faces in the graph G respectively.

Proof. Following the method used in [12], we shall begin by considering trees. We know that we only have the infinite face in a tree, so Euler's formula becomes $v(G) - e(G) = 1$. Then using proof by induction, we have the base step of $e(G) = 0$, which gives $1 - 0 = 1$ and so holds true.

Now suppose that it holds true for trees with n vertices and consider a tree T with $n + 1$ vertices. Let L be a leaf in T , so $\deg(L) = 1$. Then if L and its corresponding edge are removed, a tree T' with n vertices has been found. Putting this back into Euler's formula gives $e(T') = n - 1$, so T must have $e(T) = n$. Altogether for T then, we have $(n + 1) - n = 1$ and so the formula holds for trees.

Again we use induction to prove the formula for connected planar graphs. We have the base step $e(G) = 0$ and as we need a connected graph, this leaves us with $v(G) = 1$, and just the infinite face so $f(G) = 1$. Giving us, $1 - 0 + 1 = 2$, so the base step holds.

Let us now suppose it holds true for a graph with n edges and consider a graph G with $n + 1$ edges. Then we have two cases to prove.

Case 1. G is acyclic and the added edge does not create a cycle. In this case, it remains a tree and we know Euler's formula holds for trees.

Case 2. G contains one or more cycles. If we take an edge e out of a cycle, we create a graph G' which has one less face. Opening up the cycle merges two faces into one.

Then G' has one less edge, and one less face than G , but the same number of vertices. Substituting this into the formula gives,

$$v(G) + (f(G) - 1) - (e(G) - 1) = 2$$

which becomes $v(G) + f(G) - e(G) = 2$ and so the formula holds.

□

3.4.1 Problems

In this, our final problem section, we will see an example of how graph theory concepts can be applied to real world problems and then conclude by solving the Königsberg problem.

The first problem we will look at relates to graphs with weighted edges. Known as the *travelling salesman problem* [3, p.51], the problem is thus, what route would allow a travelling salesman to visit all of his required destinations only once and return home using the smallest amount of travelling?

Let us denote the routes using a weighted graph (G, w) , and let the weight in this case represent how much traveling is involved between each stop. As the salesman only visits each destination once, what the problem really asks is for us to find the smallest weighted Hamiltonian cycle within G .

It is sufficient to consider complete graphs for this problem, as any non-adjacent vertices within examples that are not complete can just be thought of as having a joining edge with an overly high weight, and so that edge would not be involved [3, p.51].

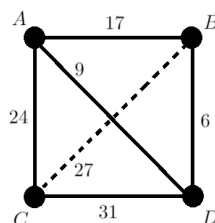


Figure 3.27: Let us consider this weighted K_4 graph and its potential Hamiltonian cycles in the table below, using A and the start and end point.

Hamiltonian Cycle	Weight
ABCD	84
ABDC	78
ACBD	66
ACDB	78
ADBC	66
ADCB	84

So for this example, the salesman's best path is $ACBDA$ or $ADBCA$ with a weight of 66. The graph's total weight is 114, $(G, 114)$, so this is a very efficient path for him.

We can deduce that the maximum number of Hamiltonian cycles a graph has is $(n - 1)!$, where n is the order and ≥ 3 (as $n \leq 2$ are trivial cases). This is because the initial vertex is fixed so the number of permutations of the remaining vertices is $(n - 1)!$. As there are pairs of cycles that are the reverse of one another and equal in weight, we can take this to be $\frac{(n-1)!}{2}$ for the number of potential weights for Hamiltonian cycles in the graph. At most, for each n we have the number of potential weights being: $n = 3 \rightarrow 1$, $n = 4 \rightarrow 3$, $n = 5 \rightarrow 12, \dots, n = 15 \rightarrow 4.35891456 \times 10^{10} \dots$

We can see that very quickly it becomes a problem that cannot be done by hand, even for a graph with a small n finding all the Hamiltonian cycles would be very time consuming. With the use of a computer, the time it will take to

calculate still becomes unreasonable, for $n = 15$ it would take over a day to compute, and $n = 20$ would take over a million years [22]. It is for this reason that the travelling salesman problem has become a computer science problem, and there is still no general algorithm available for it. There have been attempts to find optimal solutions, but these can involve special cases such as when the weights satisfy the triangle inequality [3, p.191], and again require a computer.

Now we will cover the theorem that resulted from Euler's consideration of the Königsberg problem.

Theorem 5. *A connected graph is eulerian if and only if it has no vertices with an odd degree [4, p. 51].*

Proof. Following a method of proof from [19], we first will show that it is necessary for all vertices to have even degree.

Let graph G be eulerian and T represent a Euler tour of G . Also, let the start and end of the tour be denoted by vertex u .

Then all the other vertices v within T are incident to two edges every time they are visited in the tour. These are, the edge leading to v during this part of the tour, and the edge leading out. T must contain each edge of G , so we know that all vertices are also included, due to requiring a connected graph.

Therefore, the degrees of our v vertices within T are even, double the number of times vertices are included in T . We also have u , which is incident to an edge as the tour starts and again as it ends, so also has an even degree. Therefore, if G contains a Euler tour it also has all vertices of even degree.

Using proof by induction, we now need to show that having all vertices with even degree is sufficient proof that G is eulerian.

For the base step we have $e(G) = 0$, then we only have one vertex of degree 0 and the theorem holds true. Now assume that it holds true for all G with $e(G) = n$ and all vertices of even degree.

Consider a graph G with $e(G) = n + 1$ and all vertices of even degree. As we must have all vertices of even degree, G does not have a pendant vertex and

therefore it is not a tree and will contain a cycle. Let T_1 denote the largest closed trail created by a cycle in G . Assume that T_1 is not a Euler tour and so $e(T_1) \neq e(G)$. Then the removal of T_1 's edges will create the subgraph $H = G - e(T_1)$ which is made up of one or more components.

As the removal of $e(T_1)$ will result in an even number of edges being removed from the vertices in the cycle, the vertices in H will all still have even degree. So we have that the components of H are connected with even degree on each vertex, and by our induction hypothesis they will all contain a Euler tour. Let H_1 denote a Euler tour in a component of H .

Then, as G is connected, there will be a vertex v that is present in both H_1 and in T_1 . Then let T_2 be the closed trail $T_2 = T_1 + H_1$ that can be formed through their shared v vertex. So we have $|T_2| > |T_1|$, giving us a contradiction to T_1 being the longest tour in G .

The only way to avoid this contradiction is for T_1 to include all edges of G and so be a Euler tour, therefore G is eulerian if all of its vertex degrees are even.

□

This theorem lead Euler to conclude, that it not possible to take a walk around Königsberg without crossing a bridge more than once, as the graph of the problem has all vertices with odd degree.

Corollary 5.1. *A connected graph is a Euler trail if it has, at most, two vertices of odd degree [4, p. 52].*

Proof. For graph G , lets say it contains a Euler trail, beginning at vertex u and ending at vertex w .

Then, if we connected u and w with an edge e , to get graph H , then we would have a graph with a Euler tour. Therefore, all vertices would have an even degree as per Theorem 5. It is evident then, that removing e to return to graph G , leaves us with u and w having an odd degree.

□

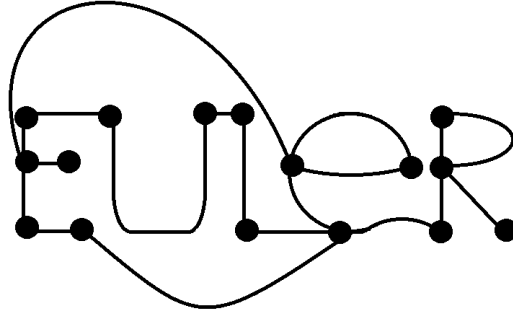


Figure 3.28: Two vertices with an odd degree can be found in this graph, one in the ‘E’ and the other in the ‘R’. All the other vertices have an even degree, so a Euler trail exists in this graph, beginning and ending at the vertices with odd degree, by Corollary 5.1. If an edge was added to join these two, then we would be able to find a Euler tour.

This concludes our formalisation of the graph theory concepts seen in Chapter 2, and the introduction of further concepts in this chapter. We will now use these as reference as we delve into the more complex ideas needed to solve the four colour theorem.

Chapter 4

The four colour theorem

The discussion in the previous chapter has laid out the groundwork to help understand and better explain the ideas we will need in this one. One topic that we are yet to rediscover and explore fully is that of planar graphs, and we are yet to see how graphs can be coloured. Both of these are vital for the four colour theorem, and so we will further investigate these ideas now.

4.1 Planar graphs

We have already seen that a graph that can be drawn so that the edges do not overlap at any point is **planar**. It is clear to see that this is a necessary requirement for the four colour theorem as it implies a ‘flatness’, which we need for a map. If we think back to our Europe map examples, having an overlapping edge would be equivalent to saying that a country that should not be able to be adjacent to another actually is, imagine that Italy lifts the toe of its boot to touch Germany.

Graphs that are planar do not have to be drawn in such a way that their edges do not cross, but when they are drawn like this then we have the **planar embedding** of the graph.

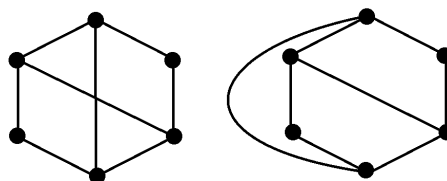


Figure 4.1: A planar graph beside its planar embedding.

The planar embedding of a graph G is denoted $\tilde{G}$, and as they have the same structure we call the edges in $\tilde{G}$ **lines**, and the vertices **points** so that we can make a distinction between the two [3, p.244].

Let us recall Lemma 1, whose proof we derived in Chapter 2, and state more clearly the two proofs that were gained from it.

Theorem 6. *A simple connected planar graph G , with $v(G) \geq 3$, must contain a vertex with $\deg(v) \leq 5$.*

Theorem 7. *A simple connected planar graph G , with $v(G) \geq 3$, satisfies*

$$e(G) \leq 3v(G) - 6.$$

We can use this to determine whether a graph is planar or not, although it is not sufficient as we shall see.

Corollary 7.1. *The graph K_5 is not a planar graph.*

Proof. Graph K_5 has $v(K_5) = 5$, and $e(K_5) = \frac{5 \times (5-1)}{2} = 10$. Using Theorem 7 we get, $10 \leq 3 \times 5 - 6 = 9$, and so we have a contradiction.

□

Recall the three utilities problem, where we proved that $K_{3,3}$ is not planar. Combined with Corollary 7.1, this leads us to **Kuratowski's Theorem**.

Theorem 8 (Kuratowski's Theorem). *A graph G is planar if and only if it does not contain a subgraph that is a subdivision of the K_5 and $K_{3,3}$ graphs.*

Before we can discuss the proof we first need to discover some new ideas, and recap others, that are required for it.

Firstly, we have already seen the concept where we can **split** a vertex into two and add an edge joining the new vertices in Chapter 3. If we flip that idea, and instead split an edge in a graph G and add a vertex to join it together again, then we have applied **edge subdivision** and the graph formed from this is the **subdivision** of G . We can think of this as performing edge deletion and then

adding a path of length 2 which begins and ends at the vertices that were adjacent to the removed edge [25].

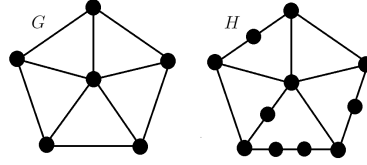


Figure 4.2: Graph H is graph G with five edge subdivisions, and so is a subdivision of G .

We can see that adding these subdivisions does not alter the shape of the graph, and so a subdivision of a graph G is planar if G is planar, and vice versa.

Lemma 9. *If, and only if, every subdivision of G is planar then G must also be planar. Conversely, if a nonplanar subdivision of G exists then G is not planar.*

Similarly, we can clearly see that a subgraph taken from a planar graph is also planar. It is evident from the planar embedding of a graph that taking a subgraph would not cause any edges to overlap and alter its planar property. Inversely, if a subgraph is nonplanar then its supergraph will be nonplanar also, no alterations have occurred to change the planar property of the subgraph.

Lemma 10. *The subgraphs of a planar graph G are also planar. Supergraphs containing nonplanar subgraphs are nonplanar.*

The use of K_5 and $K_{3,3}$ in Kuratowski's Theorem are important as these are the simplest regular nonplanar graphs. Graph K_5 is the nonplanar graph with the smallest amount of vertices, as a single vertex deletion would make it planar. Graph $K_{3,3}$ has the smallest amount of edges, as a single edge deletion would make it planar [1, p.3]. Therefore, it is sufficient to consider these two as other nonplanar graphs will be supergraphs or subdivisions of them.

This idea of a nonplanar graph that is the most simple, means that all of its proper subgraphs are planar. The nonplanar graph that has this property is called a **minimal nonplanar graph**.

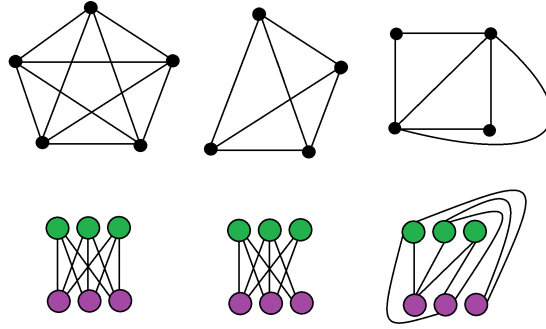


Figure 4.3: Here we see K_5 with a vertex deletion, then drawn as a planar embedding. Then the same for $K_{3,3}$ with an edge deletion. This shows us that planar graphs can be found for K_5 and $K_{3,3}$ following a single vertex or edge deletion, respectively, and so they are the simplest forms of nonplanar graphs. They are both minimal nonplanar graphs.

Next we will recap a couple of definitions from Section 3.3. A graph G is k -connected if it needs k vertices removing to increase the number of components in G . A block is a subgraph of G that cannot be broken into more components by removing a vertex, i.e. it has no cut vertices, and so is 2-connected.

By Corollary 7.1 we showed that K_5 is not planar. If we applied one edge subdivision to K_5 we would have one extra vertex and one extra edge. Using Theorem 7 again would give us $11 \leq 3 \times 6 - 6 = 12$, which might imply that it is a planar graph. However, as we have seen that a subdivision of a nonplanar graph is still nonplanar, we now know that this formula is not always sufficient, and this is where the necessity of Kuratowski's Theorem lies. Combining all the above ideas intuitively lets us see that Kuratowski's Theorem holds true, however the difficulty with the theorem is showing that it is not only necessary but is also sufficient proof of nonplanarity to say a graph contains a subdivision of K_5 and/or $K_{3,3}$.

We need one final concept before getting to the main proof.

Lemma 11. *If G is a 2-connected graph, then G contains a cycle that includes u and v , for every pair of vertices u and v .*

Proof. Let u and v be adjacent vertices in a graph G , then as we have a 2-connected graph neither can have degree one or we would be able to remove

one vertex and have a disjoint graph (part A in Figure 4.4). So another vertex must be adjacent to u and v , let us consider w adjacent to u (B). Then if we were to remove u from graph G , G will still be connected and so there will still be a path from v to w (C). So we can see that the re-addition of u would join the ends of this v, w -path in G creating a cycle that includes u and v .

□

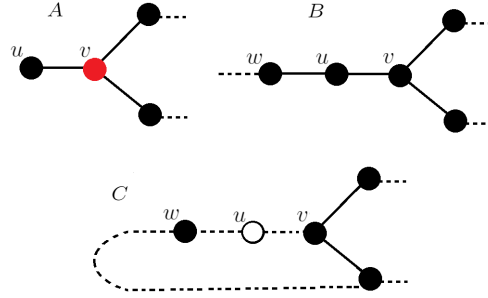


Figure 4.4: Red indicates a cut vertex and highlights how we cannot have a vertex such as u in this case with degree one. Dashed lines indicate potential paths not just edges.

We can now move onto the overall proof for Kuratowski's theorem.

Proof. Kuratowski's Theorem. Following the method in [25], we consider a proof by contradiction using a minimal nonplanar graph G that is not K_5 or $K_{3,3}$. The proof depends on three parts which we will cover and prove now.

Part 1. Show that G must be 2-connected.

We know that a graph is made up from its blocks, and from Lemma 10 we know that a nonplanar graph will have a nonplanar block. As we are considering a minimal nonplanar graph if we were able to remove one vertex and get components that include a nonplanar graph, then we have found a graph smaller than G that is nonplanar and so have contradicted the minimality of our G . Therefore, G itself must be a block to be minimal, so it cannot be 1-connected and must be 2-connected.

Part 2. Show that no vertex in G has degree 2.

Using proof by contradiction, let a vertex u in G have degree 2 and be adjacent to v and w . Then if v and w were also adjacent (they all form a triangle), and we remove u to get graph $G - u$, then as G is minimal $G - u$ will be a planar graph. If we then add vertex u back into the graph, we can see that it remains

planar in the planar embedding and so G is planar to begin with, contradicting its minimal nonplanar status.

In the case that v and w are not adjacent, let's remove u and add an edge $\{vw\}$. Again, by minimality this would be planar as we have removed a vertex. We can then subdivide the edge $\{vw\}$ to create vertex u again, and by Lemma 9 this will also contradict G being a minimal nonplanar graph.

So we have a contradiction in both cases where we can find a vertex with degree 2, so our minimal nonplanar graph cannot have a vertex of degree 2.

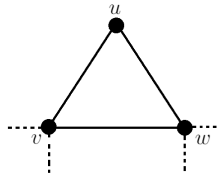


Figure 4.5: We can see from the planar embedding of a vertex with degree 2 that its addition does not change the planar property of a graph. The minimal nonplanar graph G has $\delta(G) \geq 3$.

Part 3. Show that there exists an edge e in G such that $G \setminus e$ remains a 2-connected graph.

By contradiction let's assume that G is 1-connected and let e be the edge between vertices u and v . From Part 2 we have that $G \setminus e$ would leave $\delta(u) = \delta(v) \geq 2$, and all other vertices remain $\delta(G \setminus e) \geq 3$.

Then as we know graph G is a block from Part 1 it will not contain any cut vertices. So the only way for a vertex in $G \setminus e$ to increase the quantity of components in $G \setminus e$ is for it to have degree 1 following the edge deletion. As 2 is the smallest degree in $G \setminus e$, this cannot occur and so $G \setminus e$ is 2-connected.

We will now combine these 3 parts to find the proof of Kuratowski's Theorem.

Let us now use graph $H = G \setminus e$, where e is the edge connecting vertices u and v . We know from the minimality requirement of G that H is planar, and from Part 3 that it is 2-connected. Combined with Lemma 11, this tells us that there is a cycle C in H that includes vertices u and v .

We now need to consider a planar embedding of H such that this cycle C has the maximal amount of faces inside of it, and that any other cycle that includes u and v too has the same amount of contained faces or fewer.

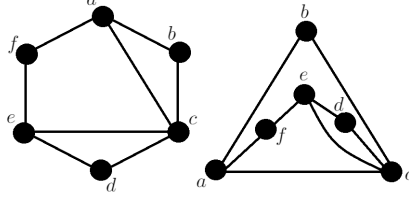


Figure 4.6: For example, these are two different planar embeddings of the same graph. We can see that the graph has 3 faces (excluding the infinite face) so 3 would be the maximal amount of faces we could fit into a cycle. For the cycle $\{a, b, c, a\}$ in the first graph only 1 face is contained, while the second puts the cycle on the outside giving the maximal amount of 3 faces inside of it.

It is evident that for C to have our maximal property that it will form the external edges of H . Let $P_1 = \{u_0, u_1, u_2, \dots, u_{n-1}, u_n\}$, where $u_0 = u$ and $u_n = v$, denote our path from u to v . The number of entries in the path is denoted by n , excluding u_0 . To complete the cycle back to u add the path $P_2 = \{u_n, u_{n+1}, \dots, u_0\}$.

As its existence would add an extra face, and so contradict the maximal requirement of C , we know that there cannot be a path that is external to C that connect vertices within the path P_1 and/or the path P_2 . Consider an edge drawn from a to e on the left hand graph of Figure 4.6, if the planar embedding had a maximal cycle that included f , drawing it external would add an extra face that would not be contained in the said cycle. We could instead draw this edge internal, to form a triangle inside the hexagon, which would preserve the maximal property of C . In our general terms, we could not have a path from u_0 to u_{n-1} , for example.

Going forward, figures in this proof will be simplified for ease of illustrating necessary ideas. Lines could indicate single edges or paths.

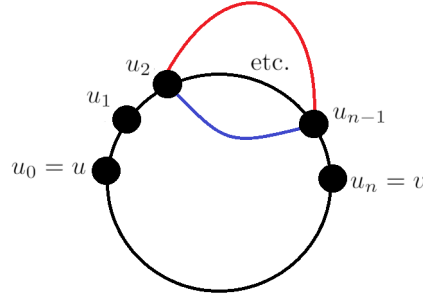


Figure 4.7: The black edges represent our cycle C . The red line represents the edge or path that cannot exist as it would add another face, while the blue path is perfectly viable here.

As $H = G \setminus e$ we know that we are missing the edge between u and v . If we could re-add this edge e in such a way as to maintain the planar embedding of H then G would be planar and we have a contradiction. So there must be an exterior path that prevents us from creating this planar embedding.

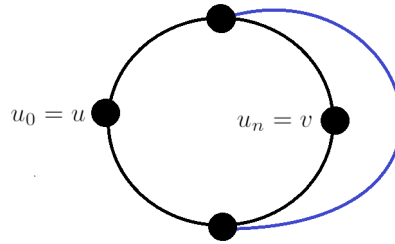


Figure 4.8: The blue line represents the path that would prevent us from being able to draw the edge e externally and maintain a planar embedding, as desired.

We now need to find and consider the cases where the paths and edges inside C prevent us from connecting u and v in a planar embedding, leading us to see the potential graphs that are nonplanar when e is added back into H to get graph G .

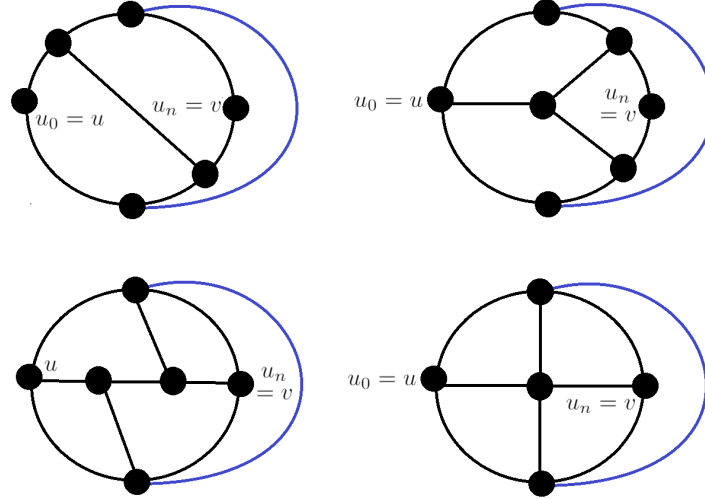


Figure 4.9: These are the potential structures that can prevent the addition of edge e forming a planar graph. The top right graph could have its inner ‘Y’ stemming from another vertex, v for example, these alternatives have the same result and so it is only drawn this one way here.

Now we have all the potential forms that a minimal nonplanar graph could take. It can be seen, through adding e back into these graphs to get back to our graph G , that they are all subdivisions of either $K_{3,3}$ or K_5 . Therefore, G does contain $K_{3,3}$ or K_5 and we have our contradiction. A nonplanar graph must contain a subdivision of $K_{3,3}$ or K_5 and Kuratowski’s Theorem holds true. □

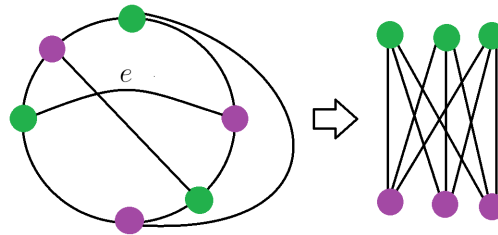


Figure 4.10: The top left graph of Figure 4.9 becomes $K_{3,3}$ when edge e is re-added. We can see that it can be split into two sets (green and purple) and each vertex in one set is connected to every vertex in the other.

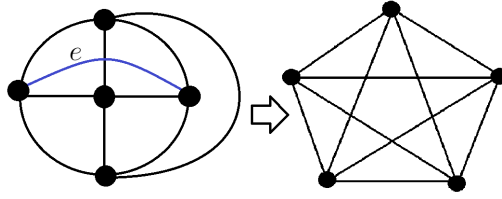


Figure 4.11: The bottom right graph becomes K_5 with edge e re-added, we can see that every vertex is connected to every other one.

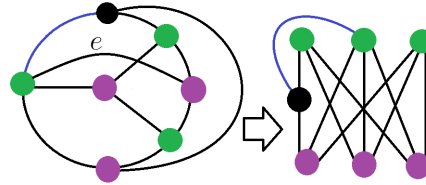


Figure 4.12: The top right graph translates to $K_{3,3}$ with edge e re-added. The black vertex is part of a path that forms a subdivision of $K_{3,3}$. The blue line is not needed as we get the subdivision of $K_{3,3}$ without it.

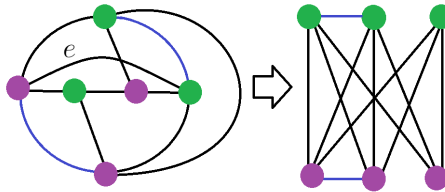


Figure 4.13: The bottom left graph becomes $K_{3,3}$ with edge e re-added. The blue lines are not needed as we get the subgraph $K_{3,3}$ without them.

This ends our discussion of Kuratowski's theorem, and we will move onto our last point in the planar graphs subsection, which wraps up an idea we saw but did not prove back in Chapter 2. Let us recall that a **maximal planar** graph is one which contains the maximum amount of edges it possibly can whilst remaining planar. In the initial proof of Theorem 7, we used the property that each maximal planar graph face is bounded by three edges. This is known as a **triangulation**, when each face has degree three.

Theorem 12. *A maximal planar graph is a triangulation.*

Proof. For a simple connected graph G with $v(G) \geq 3$, assume that it includes a non-triangulated face. Then we must have three vertices, a, b , and c that exist in

$v(G)$, that are adjacent to one another in the following way:

$$\{ab\} \in e(G), \{bc\} \in e(G), \{ca\} \notin e(G).$$

We have that $\{ca\}$ cannot exist in the edge set as this would form triangulation.

Then we can add the edge $\{ca\}$ into G and it still will be planar, showing that G was not maximally planar. Hence any face that is non-triangular can successfully have edges added until it is triangular. Repeatedly applying this notion to any non-triangular face of G would leave us with a maximal planar graph with every face triangulated [10].

□

4.2 Colouring

We have already seen examples of graphs with colour applied to them, the bipartite graphs for example have had their two vertex sets assigned different colours. The number of colours that we apply to a graph's vertices is denoted with k and we say that a graph G has a k -vertex-colouring or simply a k -**colouring**. If we have no adjacent vertices with the same colour then we have a **proper** colouring and instead can say it is k -**colourable** [3, p.288]. In this way we can see that a 2-colourable graph is also a bipartite graph. The only way to get a 1-colourable graph is for it to be an empty graph.

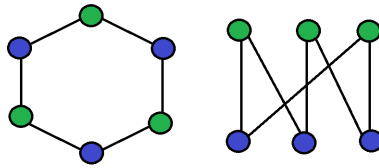


Figure 4.14: Here we can see that C_6 has a proper colouring with 2 colours and so it 2-colourable. It is then redrawn to highlight its bipartite property.

Following a similar definition, we also have k -**face colouring**, where the faces of a graph are coloured with k different colours. If no adjacent faces share the same colour then we have a proper k -face colouring and call that graph k -**face colourable**.

Finally we have **k –edge colouring**, where the edges of a graph are coloured with k different colours. If no adjacent edges share the same colour then we have a proper k –edge colouring and call that graph **k –edge colourable**.

We can convert a face colouring into a vertex colouring by finding the **dual** of a graph G . The dual G^* is formed by taking a face from G and treating it like a vertex in G^* . If the faces are adjacent in G then they have an adjoining edge in G^* .

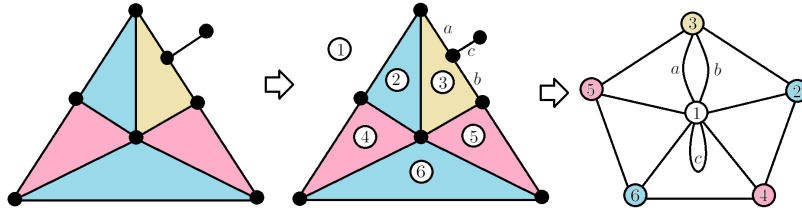


Figure 4.15: On the left we have G which is 4–face colourable, the center image shows how labeling each face helps to build the dual, and on the right we have G^* . When a face is adjacent to another face multiple times, we draw parallel edges, for example here the infinite face 1 is adjacent to face 3 at edges a and b . If a face is incident to itself, which face 1 is here as both sides of edge c count as being adjacent to the face (see Section 3.1), then we draw a loop. A loop in G^* would therefore indicate that G contains a cut vertex.

The smallest amount of colours that are required to make a graph k –colourable is that graph’s **chromatic number**, denoted $\chi(G)$. If $\chi(G) = k$ then the graph is **k –chromatic** [3, p.358]. Returning to Figure 4.14 we can see that C_6 is 2–chromatic. In fact, all circular graphs C_n , with order n , will be 2–chromatic if n is even, making them bipartite. If n is odd, then the circular graph will be 3–chromatic.

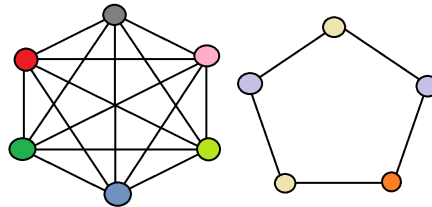


Figure 4.16: K_6 must be 6–chromatic due to every vertex being adjacent to all others and so requiring a different colour each. All complete graphs K_n are n –chromatic this way. C_5 is an odd cycle and so is 3–chromatic.

Let H be a proper subgraph of G , that is $H \subseteq G$ and $H \neq G$. If for any H we have that $\chi(H) < \chi(G)$ we say G is a **critical graph** [3, p.366]. A graph G that is k -chromatic and critical is k -**critical**.

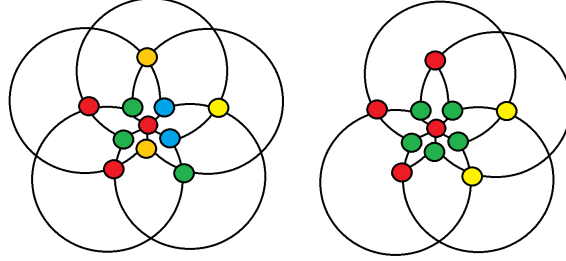


Figure 4.17: The Grötzsch graph, seen on the left, was discovered by Grötzsch in 1958 and Mycielski in 1955. It forms part of a series of graphs where the next graph in the series has a larger chromatic number by one than the previous and all graphs include no triangles (cycles of length 3) [3, p.366]. The Grötzsch graph is fourth in the sequence so is 4-chromatic, it is also critical and is therefore a 4-critical graph. We can see that removal of one edge has allowed us to find a 3-colouring.

Theorem 13. *If G is a k -critical graph then every vertex in G has $\delta(G) \geq k - 1$.*

Proof. Using proof by contradiction let graph G have a vertex $v = \delta(G)$ with $\delta(G) < k - 1$. Then as G is a critical graph, $G - v$ is $(k - 1)$ -colourable.

Graph G can be partitioned into vertex sets V_i , where each set contains the vertices that are coloured with the colours $1 \leq i \leq k - 1$ in the following way:

$$\{V_1, V_2, \dots, V_{k-1}\}.$$

Then, as $\deg(v) = \delta(G) < k - 1$ at least one of these sets, say V_j , contains no neighbours of v . Then it would be possible to introduce v back into the graph to form G by having it adjacent to a vertex in V_j . So we have found a $(k - 1)$ -colouring of G , which is a contradiction. It therefore must be that $\delta(G) \geq k - 1$.

□

Now that we have an understanding of colouring within graphs, we can begin to discuss related theorems.

4.2.1 Six colour theorem

Theorem 14. *Every simple connected planar graph is 6-colourable [13].*

Proof. Using proof by induction, assume that graph G has $v(G) \leq 6$. Then we can clearly find a successful 6-colouring for G . Next, assume that the theorem then holds for a graph with $v(G) = n$, and let G have $v(G) = n + 1$.

We know from Theorem 6 that a connected planar graph must have one vertex with $\deg(v) \leq 5$. Let this vertex be known as u , and consider $G - u = G'$. Then, G' has $v(G') = n$ and is assumed to be 6-colourable. As u has degree ≤ 5 we can add it back into G' to find G without stopping the 6-colouring being satisfied, as there will always be a 6th colour option available for u that it is not adjacent to.

Therefore, as the theorem holds for $n + 1$, the six colour theorem holds true. □

4.2.2 Five colour theorem

Theorem 15. *Every simple connected planar graph is 5-colourable [3, p.291].*

This theorem was first proved by Heawood, the mathematician that found a counter example to disprove Kempe's proof of the four colour theorem, which we saw in Chapter 2. The following proof was discovered at this time and utilised the idea of Kempe chains in a similar fashion to which we saw previously.

Proof. First consider a vertex v that has the maximum degree found in its graph G . If $\Delta(G) = \deg(v) \leq 4$ then a 5-colourable solution can clearly be found to satisfy v .

If $\Delta(G) = 5$ and 2 or more of its neighbours share the same colour, then again v can clearly be satisfied. It remains to consider the case where the 5 neighbours of v include all 5 potential colour choices.

Let the 5 neighbours of v be u_1, u_2, u_3, u_4, u_5 , where u_i is assigned the colour $1 \leq i \leq 5$. The first way to solve this predicament is to find the Kempe chains leading from u_1 and u_3 that use colours 1 and 3. If the two subgraphs of G that these two chains create share no vertex, then they are not connected, and it would be possible to invert the colours of one chain to free up a colour for v .

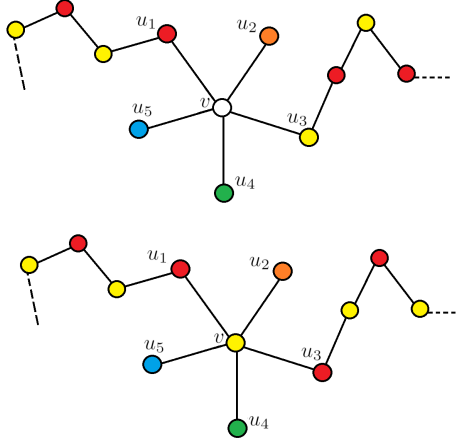


Figure 4.18: Here we see the case where the chains leading from u_1 and u_3 are not connected, and so we can invert one chain to free up a colour. Here, colour 3/yellow has been made available for v .

The next case that could arise is when the subgraphs for the chains leading from u_1 and u_3 , that use colours 1 and 3, share vertices. I.e. they belong in the same Kempe chain.

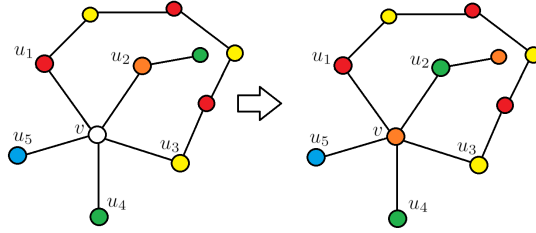


Figure 4.19: In this case, if u_1 and u_3 are in the same chain, then the chains stemming from u_2 are enclosed. These could be a chain containing colours 2 and 4 or 2 and 5, but either way, inverting the colours in one of these u_2 chains will again free up a colour for v as there is no risk of that chain being adjacent to u_4 or u_5 .

The final consideration is when u_1 and u_3 share a chain, and u_2 also shares a chain with either u_4 or u_5 . In this case, as we can see by considering the planar embedding of vertex v in Figure 4.19, this second chain would contradict the planar property of the graph G . Therefore, we have covered all of the potential planar cases, and G can be 5-colourable. The five colour theorem holds true.

□

4.2.3 Four colour theorem

Theorem 16. *Every simple connected planar graph is 4-colourable.*

We have briefly noted previously some of the concepts that were required to prove the four colour theorem. Computer usage is necessary to reach the conclusion that the theorem holds true, so in this subsection no attempt to prove it will be made but we will discuss the ideas of **unavoidable sets**, **reducibility**, and **discharging** that paved the way for Appel & Haken to prove it in 1976.

Throughout this section we will follow ideas from the [3] *Colouring of Maps* chapter. The main idea for this discussion of the proof centers around proof by contradiction. Similar to the Kuratowski's proof method we will be considering the smallest counter example of a graph G . This G is not 4-colourable and is minimal in that removal of any edge or vertex would make it so.

Furthermore, our G must be 5-critical so that every proper subgraph is 4-colourable, supporting its minimality. It follows then from Theorem 13 that $\delta(G) \geq 4$.

Proposition 17. *Let graph G have every vertex degree $\delta(G) \geq 4$, and be minimal in regards to being not 4-colourable and hence being 5-critical. Then G is a triangulation.*

Proof. Let there be a face in G that is contained by a cycle of length > 3 . Then as G is a planar graph there will be two vertices x and y that are not adjacent to one another. Then if we identified (Section 3.3) x and y into a vertex z and let the new graph be denoted G' , we have found a new graph G' that has the same number of edges and one less vertex as G , and would remain planar as evident in the graph's planar embedding.

As G is 5-critical this vertex removal results in a 4-colourable graph. If we were to assign z the colour i , then we can see that reversing the identifying of z and colouring x and y with i maintains the 4-colouring as they are not adjacent.

So we have found a 4-colouring of G , which is a contradiction, so it must only contain cycles of length 3 and be a triangulation.

□

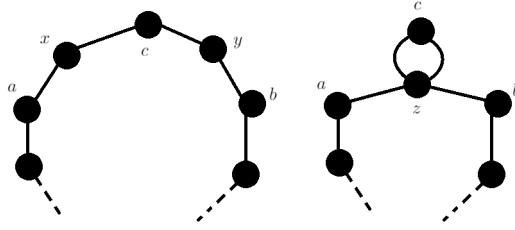


Figure 4.20: Proposition 17 visualised.

From Theorem 6 we have that a simple connected planar must have a vertex v with $\deg(v) \leq 5$. Combined with our graph G having $\delta(G) \geq 4$ tells us that this vertex v must have degree 4 or 5. To find the contradiction we must consider both cases.

Proposition 18. *Let graph G be defined as in Proposition 17, then G does not contain a vertex with degree 4.*

Proof. This proof came from Kempe and employs the use of Kempe chains. Again we begin by contradiction, let v be a vertex with degree 4 in G . As G is 5-critical we have that $G - v$ is 4-colourable. Let $\{V_1, V_2, V_3, V_4\}$ be the vertex partitions of the 4 colours used.

As G itself is not colourable v must be adjacent to all four colours, so its neighbours must be v_1, v_2, v_3, v_4 where $v_i \in V_i$ for $1 \leq i \leq 4$. Consider the subgraph of $V_i \cup V_j$.

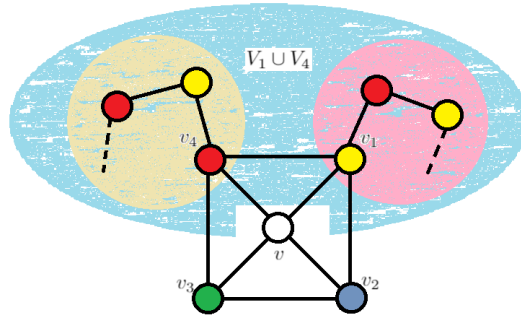


Figure 4.21: Lets consider $V_1 \cup V_4$ for this example. If v_1 and v_4 are in different components of the subgraph, as we have here shown in pale yellow and pink, then we can invert the colours in one of the components to obtain a new 4-colouring. As we have seen in previous Kempe chain examples this would result in freeing up a colour for v and would make G 4-colourable, giving us a contradiction. Hence, v_1 and v_4 or v_i and v_j need to be in the same component of the subgraph, so they share a Kempe chain.

Now let us consider these Kempe chains as paths denoted P_{ij} that connect v_i and v_j . As we have a connected graph there must be a path between every pair of vertices. With v_1, v_2, v_3, v_4 labelled clockwise as in Figure 4.21, let there be a cycle $C = \{vv_1P_{13}v_3v\}$.

Theorem 19 (Jordan Curve Theorem). *A simple closed curve C partitions a planar graph into two components that have C as their boundary [3, p.245].*

We will not attempt a proof of the Jordan Curve Theorem in this project, as despite being intuitively clear it has a very involved proof. The two components are known as the interior of C and the exterior, denoted $int(C)$ and $ext(C)$ respectively. As C is the boundary of both we have that $int(C) \cap ext(C) = C$. The theorem tells us that a path from a vertex in $int(C)$ to a vertex in $ext(C)$ will include a point in C .

Returning to our cycle C we can now apply the Jordan Curve Theorem to see that it creates its own $int(C)$ component that contains v_2 separating it from v_4 which is in $ext(C)$. We now know that the path between these vertices, denoted P_{24} , must meet at a vertex in C as we have a planar graph, call this vertex u . However, as C is a Kempe chain containing only the colours 1 and 3 it is not possible for u to be in the path P_{24} containing only colours 2 and 4. So we have our contradiction and G cannot contain a vertex of degree 4.

□

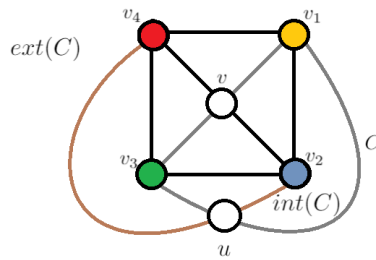


Figure 4.22: Visualising the proof for Proposition 18. Cycle C is shown in grey, and P_{24} in brown. Vertex u is the point that must exist, but cannot, by the Jordan Curve Theorem.

Proposition 18 leaves us with the case that the vertices of G have $\delta(G) \geq 5$, and the vertex v in G as required from Theorem 6 must have $deg(v) = 5$. This leads us to the following two observations.

Proposition 20. *Let graph G be defined as in Proposition 18, then G does not contain a 4-cycle that when removed separates G into two components.*

Proof. Using proof by contradiction, let the planar embedding of a graph G contain a 4-cycle and let vertex v have $\deg(v) = 5$. Let v be contained within the 4-cycle and be adjacent to none of the vertices within the cycle, so that it would be separated from G if the 4-cycle was removed. From the definition of G we require each vertex to have $\delta(G) \geq 5$, and following this condition does not allow a planar embedding.

Equally, if v was adjacent to 1, 2, 3 or 4 of the vertices of the cycle, we cannot find a planar embedding that allows all vertices to have $\delta(G) \geq 5$. Hence, we have a contradiction to the planar property of G , and a 4-cycle cannot exist in G . $\square$

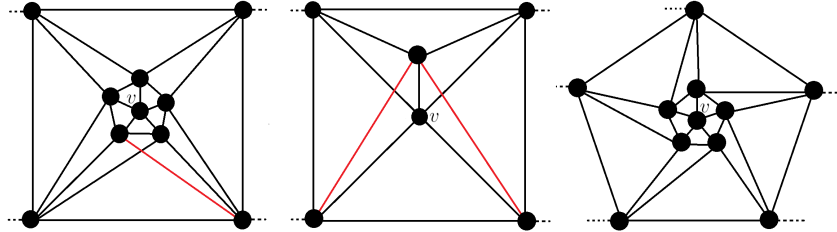


Figure 4.23: The red edges here represent the edges that are required to satisfy the degree of our vertices, yet contradict the planar property of G . The first case shows a possible graph G where v is not adjacent to any vertices within the cycle, and the second shows when v is connected to all four. On the right we have vertex v surrounded by a 5-cycle, in this case we have maintained the definition of G , and have a 5-cycle that the removal of would separate v and its neighbours from G . It is a separating 5-cycle.

Proposition 21. *Let G be defined as in Proposition 20, then G is 5-connected.*

Proof. As G has $\delta(G) \geq 5$ and is minimal so has no cut vertices, we know that removal of 4 or fewer vertices adjacent to any vertex v does not leave G disconnected, v will still have at least one other neighbour. We also know that G contains a vertex v that must have $\deg(v) = 5$. The vertex v can have all of its neighbours removed to disconnect it, and so G is 5-connected. The right hand graph in Figure 4.23 highlights an example of this case. $\square$

Expanding upon these propositions, mathematician George Birkhoff set up the definition **essentially k -connected**. This means that every separating

cycle of length $k - 1$ in a graph G stems from the neighbours of a vertex with degree $k - 1$. Let us again consider the right hand side graph from Figure 4.23.

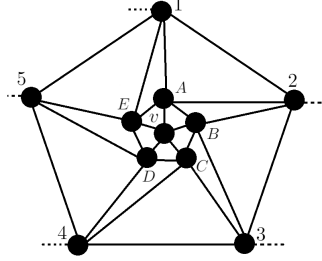


Figure 4.24: To illustrate what is meant by essentially k -connected let this be a subgraph of a graph G containing two separating 5-cycles, let the cycles be $C_1 = \{A, B, C, D, E, A\}$ and $C_2 = \{1, 2, 3, 4, 5, 1\}$. We can see that C_1 is the set of vertices that stem from vertex v with degree 5, and that C_2 is induced by the set of vertices from C_1 . If all the separating cycles of length 5 in the graph G were found as a consequence of a vertex of degree 5 then G would be essentially 6-connected.

Furthermore, Birkhoff demonstrated that a minimal counter example to the four colour theorem must be essentially 6-connected. The proof comes from combining the triangulation property of G and Proposition 21. It is evident that to disconnect a vertex of degree 5 in G using 5 vertices that we need to remove all of its neighbours, and as they need to form triangles they will be adjacent and so will form a separating 5-cycle.

Proposition 22. *Let G be defined as in Proposition 21, then G is essentially 6-connected.*

Kempe's failed proof.

We will now take this moment to recap and formalise the ideas we saw in the History chapter in relation to Kempe's attempt at a proof. Let us consider the Europe maps we saw previously as if we have taken their dual and instead see the more traditional graph form we are now used to.

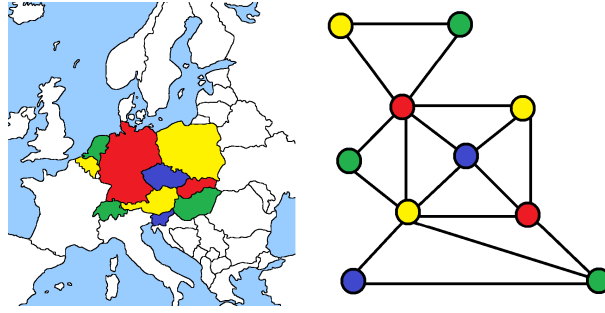


Figure 4.25: An example of transforming our previously considered Europe maps into a graph by taking its dual.

We now know that the ‘proof’ failed in the case where Kempe considered a face that was surrounded by 5 countries. Kempe, in essence, was considering a minimal graph G as defined in Proposition 22, with the face being the vertex v that must have degree 5.

By the minimality of G the graph $G - v$, where $\deg(v) = 5$, has a 4-colouring. Kempe’s idea was that $G - v$ can be manipulated via the use of Kempe chains to allow the successful re-entry of v , which would mean G was also 4-colourable; providing a contradiction and so giving the conclusion that there is no minimal counter example to the four colour theorem and so the theorem holds true.

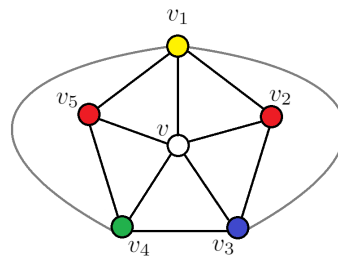


Figure 4.26: Kempe’s argument illustrated.

In Figure 4.26 we have vertex v with degree 5 and the grey lines representing Kempe chains containing the colours of the vertices they link to. Let yellow be colour 1, red 2, blue 3 and green 4. It is assumed that the Kempe chains from v_1 and v_3 are in the same component of the subgraph G_{13} (where the subscript denotes the colours in the subgraph), otherwise we could invert the colours of one component to free up a colour for v . The same argument holds for v_1 and v_4 in G_{14} .

As we have previously done let there be a cycle $C_1 = \{vv_1P_{13}v_3v\}$ and another $C_2 = \{vv_1P_{14}v_4v\}$, where P_{13} and P_{14} are the paths of the Kempe chains. We can see that C_1 separates v_2 and v_4 , and C_2 separates v_3 and v_5 . Therefore, v_2 and v_4 are in separate components of G_{24} , and v_3 and v_5 are in separate components of G_{23} .

This led Kempe to observe that we could invert the red/blue chain of v_5 as it does not affect v_3 , and invert the red/green chain of v_2 as it does not affect v_4 . This leaves v_5 blue and v_2 green, leaving only 3 colours in the 5-cycle surrounding v so that v can be successfully given a colour.

Heawood disproved this by finding a counterexample where the chains from v_5 and v_2 are adjacent at a point that does not include their shared colour. For example, we have blue/red from v_5 and red/green from v_2 so if a blue vertex in v_5 's chain was adjacent to a green one in v_2 's chain, when they were both inverted we would end up with red and red being adjacent.

Overall, despite being proved wrong, Kempe did pave the way for the final successful proof. His methods introduced the ideas of reducibility and unavoidable sets that are vital going forward. Both of these require the definition of a **configuration**.

If we have a cycle C in a simple planar graph G and an edge that is in $e(G)$ but not in $e(C)$, and this edge begins and ends on vertices within C , then the edge is known as a **chord** [3, p.144]. A configuration is a cycle C that has no internal chords and contains a triangulation. The cycle C is known as the **bounding cycle** of the configuration.

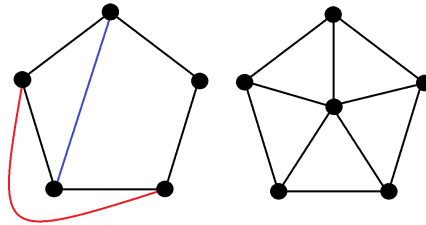


Figure 4.27: Let these two graphs be subgraphs that contain a 5-cycle from a graph G . On the left we have a red external chord and a blue internal chord, and so this subgraph is not a configuration. The right hand graph contains a triangulation, and no internal chords, so is a configuration and its separating 5-cycle is a bounding cycle.

In this way we can see that wheel graphs W_n with $n \geq 2$ vertices are configurations.

A configuration is **reducible** if it cannot be found in the smallest counter example to the four colour theorem, i.e. the graph that we have been considering as defined in Proposition 22. For example, W_4 is reducible as it has a vertex, its hub, with degree 4 which cannot exist by Proposition 18. One well known example of a configuration is the Birkhoff diamond.

Theorem 23. *The Birkhoff diamond is reducible.*

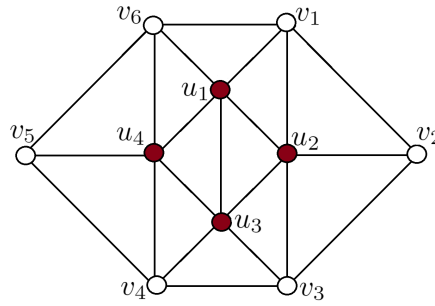


Figure 4.28: The Birkhoff diamond graph.

Proof. Let the Birkhoff diamond be as illustrated above, and be a configuration of a graph G that has the properties as defined in Proposition 22. The vertices $\{v_1, v_2, v_3, v_4, v_5, v_6\}$ then form a bounding cycle of this configuration. The vertices $\{u_1, u_2, u_3, u_4\}$ all have degree 5 and form a triangulation.

To discover if there is a way to 4-colour this configuration we begin by ‘simplifying’ it. Let this simplified graph be denoted G' . Graph G' is formed by removing the internal vertices of the bounding cycle, then identifying the vertices v_1 and v_3 to create $v_{1,3}$, and removing one of the edges that now form a parallel pair between $v_{1,3}$ and v_2 . Then add a connecting edge between $v_{1,3}$ and v_5 .

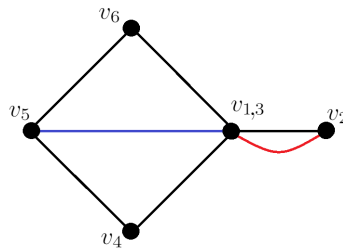


Figure 4.29: Graph G' with the red edge indicating where the parallel edges would be after identifying v_1 and v_3 , and the blue indicating the new edge added into G' .

We are allowed to simplify G in this manner as we know by Proposition 22 that no two non-adjacent vertices in the bounding cycle have an adjoining edge. If they did a separating 5-cycle would be found that is not induced by a vertex of degree 5, contradicting G being essentially 6-connected. Therefore, we can identify non-adjacent vertices in the bounding cycle as we know there will be no conflict in giving them the same colouring later on.

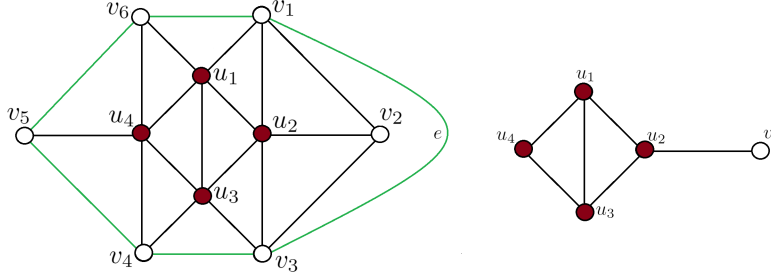


Figure 4.30: Edge e is the edge that cannot exist or the green separating 5-cycle would be created. The component it separates is shown on the right, and as it contains v_2 that does not have degree 5 this contradicts the essentially 6-connected property of G .

The creation of G' benefits us as if we were to consider the potential 4-colourings of the bounding cycle of length 6 without breaking it down we would find 31 different cases, not counting permutations of colours [34, p.180].

If we have a case such that a 4-colouring can be found for vertices $\{u_1, u_2, u_3, u_4\}$ upon re-addition into the graph, without needing to alter the colours on the bounding cycle via Kempe chains, then we have found a **good colouring**. It can be shown that all other cases lead to a 4-colouring but require Kempe chains inverting to put them into the same form as a good colouring first.

Furthermore, we can see that some of the good colourings with $\{v_1, v_3\}$ being the same colour are symmetries of others with $\{v_4, v_6\}$ being the same colour, so we only need to show one of them is reducible for the other to be also. By the creation of G' , therefore, we result in a graph that leads us to consider fewer of the cases that are not good colourings and reduces the amount of symmetrical cases that arise, making our job easier.

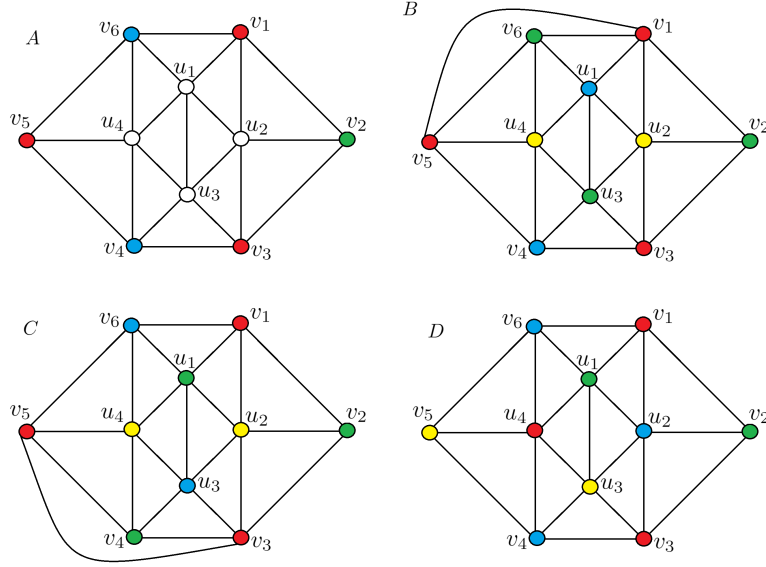


Figure 4.31: For example, we can see that the colouring of the Birkhoff diamond in *A* is not a good colouring, there is no way to successfully colour the inner vertices, but in the following three ways we can make it one. Firstly, if we had a red-yellow Kempe chain from v_1 to v_5 , as in *B*, then we could invert the blue-green chain from v_6 to find a good colouring. Similarly, in *C* we can invert the blue-green chain of v_4 , if we have a red-yellow chain from v_3 to v_5 . If neither of these options exist then it must be that the red-yellow chain from v_5 can be inverted without affecting any other vertex in the bounding cycle, so turning v_5 yellow and finding a good colouring as in *D*.

Returning to graph G' we see that it has fewer edges and vertices than G , and so by G' 's minimality it must be that $G' \setminus \{u_1, u_2, u_3, u_4\}$ is 4-colourable. This leads us to the following observations of how a successful colouring of G' could be achieved:

- v_1 and v_3 receive the same colour
- v_5 is a different colour to the one taken by v_1 and v_3
- v_2 is a different colour to the one taken by v_1 and v_3
- v_4 and v_6 are different to the one or two colours assigned to v_1, v_3 and v_5

When all the potential colourings are considered of this format we are left with 5 cases. Permutations of colours (say the cycle was coloured with colours {red, blue, red, green, blue, green} from v_1 to v_6 , this would be equivalent to being coloured {yellow, green, yellow, blue, green, blue} and so we do not need to consider it twice), colourings that are essentially the same ({red, blue, red,

blue, red, green} and {red, blue, red, blue, red, yellow} for example would be considered the same), and symmetries are not included in this count. Let us denote the four colours as 1, 2, 3, 4 and the potential colourings c_i for $1 \leq i \leq 5$, then the cases are:

	v_1	v_2	v_3	v_4	v_5	v_6
c_1	1	2	1	3	2	3
c_2	1	2	1	4	2	3
c_3	1	3	1	4	2	3
c_4	1	3	1	4	2	4
c_5	1	3	1	3	2	3

Cases c_1, c_2, c_3, c_4 are all good colourings. While c_5 is not a good colouring we can see that its format is equivalent to the graph A in Figure 4.31 if we considered its symmetry, and so we have already shown that it can be made into a good colouring.

Therefore we have found 4-colourings for all cases of $G' + \{u_1, u_2, u_3, u_4\}$. This is a 4-colouring of G and so we have a contradiction and the Birkhoff diamond is reducible.

□

We can now observe that Kempe's 'proof' failed because it followed the requirement that a minimal counter example must be essentially 6-connected and so must contain W_5 , but did not correctly show that W_5 is reducible.

If we take a set $\mathcal{U}$ of configurations, we say that this set is **unavoidable** if at least one element from it must exist in our minimal counter example to the four colour theorem, graph G . For example, consider $\mathcal{U} = \{W_5\}$. The graph W_5 must be present in G by the requirement of needing a vertex with degree 5, and so this set is unavoidable, and any set containing W_5 will be.

The idea to conclude the four colour theorem proof is to show that every element in an unavoidable set, one containing all the possible configurations that must exist in a counter example, is reducible. We have already encountered Appel & Haken who made the first conclusive proof of the four colour theorem, and it was this method that they employed.

It may seem that you could work out the whether the configurations in the set are reducible or not by hand, just as we did with the Birkhoff diamond, despite how painstaking this may be. However, when we consider that the first set that Appel & Haken used included 1936 elements, and each of these elements in turn has multiple colourings to be considered, we begin to see the necessity of computers in the four colour theorem proof.

It is all the more evident when we note that while the Birkhoff diamond, with bounding cycle of length 6, only produced 31 potential colouring options, configurations with a longer bounding cycle produce many more. If the bounding cycle had length 7 this would be 91 options, a bounding cycle of length 8 has 274 options, length 9 has 820, and so on...and it was thought by mathematician Heesch that configurations with cycle length up to 18 would be required [34, p.181].

Heesch introduced **Heesch representation**, which helped to make the size of configurations more concise and therefore easier to handle by computers of the time. The Heesch representation reduces a configuration to its internal vertices only and the degree of each vertex is denoted next to it.

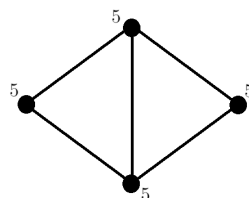


Figure 4.32: For example, this would be the Heesch representation of the Birkhoff diamond.

Alternatively, Heesch representation may include different shapes to denote a vertex, and that shape indicates the degree of the vertex. We have most commonly found that $\bullet$ is used to indicate $\deg(v) = 5$, $\circ$ implies $\deg(v) = 6$ and vertices with larger degree are denoted with a circle containing the value of their degree [33, p.125]. With this notation we could easily reverse a Heesch representation to find the original configuration.

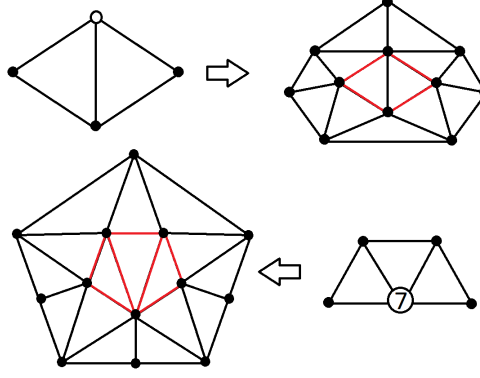


Figure 4.33: Two Heesch representations transformed back into their configurations. The red edges indicate the component that the bounding cycle separates, it is this that is used in the Heesch representation. The black dots in the configurations do not mean a degree of size 5 now, but have returned to their vertex identity.

Heesch also formulated a **method of discharging** which is used to generate unavoidable sets. The degree of a vertex plays an important role in discharging, giving further importance to Heesch representation.

The **charge** of a vertex is defined as $6 - \deg(v)$. A charge is positive for a vertex v with $\deg(v) \leq 5$, 0 with $\deg(v) = 6$, and negative for $\deg(v) > 6$. To **discharge** a graph is to remove all positive charge where possible by distributing the existing charge based on a particular **discharging algorithm**.

Theorem 24. *A graph G with triangulation has a total charge of 12.*

Proof. Recall Lemma 1 where we found that maximal planar graphs, which are triangulated by Theorem 12, have $e(G) = 3v(G) - 6$.

The total charge of G will be,

$$\sum_{v \in v(G)} (6 - \deg(v)) = 6v(G) - \sum_{v \in v(G)} \deg(v).$$

Combining Theorem 2 with Lemma 1 gives us,

$$\sum_{v \in v(G)} (6 - \deg(v)) = 6v(G) - 2e(G) = 12.$$

□

As for our counter example graph G we are considering triangulated graphs, we have from Theorem 24 that the charge in G will always be constant at 12. This tells us that we cannot fully discharge G and so a discharging algorithm will leave us with one or more subgraphs within G that have a positive charge.

Any triangulated graph G must contain some of these positively charged subgraphs, otherwise it would not be able to make up its charge of 12, contradicting Theorem 24. The subgraphs could be multiples of the same configuration, twelve W_5 's for example, or a collection of different ones. It is these positively charged subgraphs of G that form an unavoidable set, our minimal counter example graph G must contain at least one and so they are unavoidable.

The next theorem shows an example of how discharging algorithms are designed to highlight configurations that then form unavoidable sets.

Theorem 25. *A simple planar graph G with triangulation and $\deg(v) \geq 5$, finds the following two configurations after applying a discharge algorithm that states: every vertex of degree 5 must distribute its charge of 1 equally to all of its neighbours.*

(i) *an edge that begins and ends with vertices of degree 5*

(ii) *an edge that begins at a vertex with degree 5 and ends at one with degree 6*

Proof. We have a discharging algorithm that distributes the charge of every vertex of degree 5 equally to all of its neighbours.

Then the maximum amount of charge a vertex can receive is $\frac{\deg(v)}{5}$. A vertex v then has, at most, a total charge following the algorithm that equals,

$$6 - \deg(v) + \frac{\deg(v)}{5} = 6 - \frac{4}{5}\deg(v).$$

Hence, a vertex can only end up with a positive charge if it started with a degree of ≤ 7 . This comes from the inequality,

$$0 < 6 - \frac{4}{5}\deg(v) \Rightarrow \frac{4}{5}\deg(v) < 6 \Rightarrow$$

$$4\deg(v) < 30 \Rightarrow \deg(v) < 7.5.$$

We have that G only contains vertices of degree 5 or higher so only vertices of degree 5, 6 or 7 can end up with a positive charge with this particular discharging algorithm.

If a vertex v has degree 5 then it must have a neighbour u with $\deg(u) = 5$ to end up positive. Similarly a vertex v with $\deg(v) = 6$ must have a vertex of degree 5 as a neighbour to result in a positive charge. So we have an edge $\{uv\}$ that satisfies both cases.

If $\deg(v) = 7$ then as its initial charge will be -1 , it needs a minimum of 6 adjacent neighbours with degree 5 to end up positive. Since we have a triangulation, the vertices that are neighbours of v will form a cycle, and there is only room in the cycle for at most one vertex w with a degree ≥ 6 . This vertex w cannot stop all instances of vertices of degree 5 in the cycle from being adjacent. So we again must have an edge that begins and ends with a vertex of degree 5.

We have considered all the possibilities and have found that both of the required configurations must exist for there to be a positive subgraph in G . Therefore, these belong to an unavoidable set $\mathcal{U}$ that follows from this discharging algorithm, and the theorem holds true.

□



Figure 4.34: The two configurations of our unavoidable set in Heesch representation.

We will now further the observation that these positive subgraphs form the configurations of an unavoidable set with a proof that the particular set $\mathcal{U}$ from Theorem 25 is unavoidable.

Proof. Using proof by contradiction, let us consider a graph G with triangulation with the vertices of degree 5 only having neighbours of degree 7 or more.

Then any vertices of degree 5 or 6 will end up with a charge of 0 as they have no neighbours of degree 5 to receive charge from.

A vertex v with degree n , where $n \geq 7$, cannot have consecutive neighbours of degree 5, i.e. no adjacent vertices in its bounding cycle are both of degree 5. So at most v can gain $\frac{1}{2}n \times \frac{1}{5} = \frac{1}{10}n$ charge.

Vertex v started with $6 - n$ charge so now has, at most, $6 - n + \frac{1}{10}n = 6 - \frac{9}{10}n$. A vertex with degree 7 will provide the largest amount of positive charge to G , that is:

$$6 - \frac{9}{10}n \leq 6 - \frac{9}{10} \times 7 = -0.3 < 0.$$

Therefore, graph G would end up with a negative charge which contradicts Theorem 25, and so $\mathcal{U}$ is an unavoidable set.

□

It is evident that finding an unavoidable set that contains enough configurations to be a sufficient proof to the four colour theorem is going to be an arduous task. In addition to finding the set, we are still left with the task of showing that each configuration in it is reducible. It is no small wonder that it took almost 200 years for a proof to be found. Indeed, when Appel attended a lecture by Haken about the difficulty in trying to find a proof, Haken was quoted as saying ‘I’m quitting. I consider this to be the point beyond which one cannot go without a computer. [34, p.195]’ It was Appel’s offer to assist Haken with the computer work following this that kept the show on the road.

Before this, Heesch had initiated the study into computer usage in 1965 following a conversation with Haken in which Haken inquired as to whether computers would aid the progress of a proof. Heesch found that a test for reducibility could be translated well into a computer algorithm and got to work. At the time, he had access to a supercomputer which computed configurations with a bounding cycle of length 12 in 6 hours and one with length 13 in 16 – 31 hours [34, p.181]. Anything beyond this was unrealistic given the computers capabilities, and it was not feasible for Heesch to have access to the supercomputer for that length of time. It did not help Heesch’s case that he was considering a proof that involved around 8900 configurations, some with bounding cycles of length 18 which come with in excess of 18 million potential colourings, estimating a need for anything between three thousand to fifty thousand hours worth of computer time.

In 1971 Haken decided to find a new approach which would decrease the time it would take a computer to run an algorithm. Instead of searching for reducible

configurations, which was the most fashionable idea among mathematicians at the time, Haken searched for an unavoidable set [34, p.193]. While he believed that this would be more efficient, Haken felt that technology was not strong enough for what he wanted to do and debated leaving his ideas to one side to wait for it to improve. This is when he held the lecture that Appel attended and ultimately joined the team.

Appel & Haken went on to succeed where others had failed because this new approach lent itself to a realistic expectation of what a computer could achieve in the 1970's. Starting work in 1972 and concluding in 1976, they used 1000 hours of computer time to find the final unavoidable set. This unavoidable set contained all the possible unavoidable configurations for a minimal counter example, and all were proved reducible. The final submitted proof stood 4 feet of the floor when printed, and used 487 discharging algorithms to gain this set of 1482 unavoidable configurations [34, p.226].

More streamlined proofs have since been found following the improvement of technology. For example, in 1997 Robertson et al. used only 32 discharging algorithms to create an unavoidable set containing 633 reducible configurations that proved the theorem.

We have now reached the end of our four colour theorem chapter. We have seen how it can be proved and explained the concepts that are required. For our remaining objective, we will now begin looking at using graph theory and colouring ideas on other surfaces.

Chapter 5

Other surface embeddings

But what happens next? Knowing how to prove the four colour theorem is not the end of the road. It has opened up other ideas and branches of graph theory that lend themselves to solving even more complex problems, and so the discussion continues. In this chapter we will briefly see how colouring theorems are being applied to other surfaces and shapes and consider some of their properties too.

A **surface** is a connected 2–dimensional manifold [3, p.276]. While we make no attempt to explain this to any useful means in this project, consider it to mean that you have an object that at a local level appears straight forward and flat and yet at a global level it is not. It is the idea behind believing that the Earth is flat, from our viewpoint we cannot tell any different, but when viewed as a whole we can see that it is a sphere (roughly speaking). For our purposes now we can just consider the meaning to indicate that we are going to look at structures different to the 2–dimensional planar embeddings that we have been considering throughout the rest of the report.

5.1 Polyhedra

To begin with we shall consider **polyhedra**. Singularly known as a polyhedron, these are solid 3–dimensional shapes comprised of flat polygonal faces, with straight edges and vertices. The cube and pyramids are polyhedra for example.

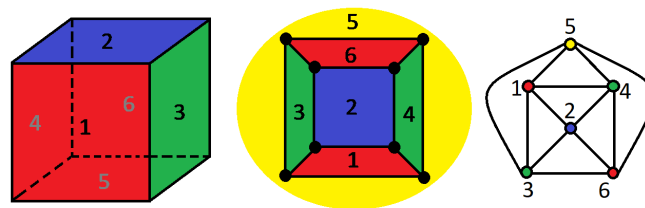


Figure 5.1: Here, on the left, we have a cube where the grey numbers indicate the faces which are not visible to us. Then we can draw the cube as a map, with the 5 face becoming the yellow infinite face. Finally, on the right we have its dual graph, showing that the cube can be represented by a planar embedding.

The central diagram in Figure 5.1 is an example of a **polyhedral graph**, these graphs have vertices and edges that represent the vertices and edges in a polyhedra, with the same incidence function [3, p.21]. In this way we can see that polyhedra are isomorphic to a planar embedding and so are 4-colourable.

From Theorem 4 we can also see that Euler's formula $v - e + f = 2$ will hold for polyhedra as this is the form it takes for connected planar graphs. For example, Euler's formula for the cube becomes $8 - 12 + 6 = 2$, and for a trigonal prism (a cheese wedge) we have $6 - 9 + 5 = 2$, as expected.

Additionally we have **regular polyhedra**. These are polyhedra with faces all of the same type and where each vertex has the same arrangement of faces surrounding it [34, p.41]. We can see that a square based pyramid is not regular, while a triangular based pyramid is. We can make use of Euler's formula to show that only 5 regular polyhedra exist, together they are known as the **platonic solids** [3, p.21].

Theorem 26. *There are only 5 platonic solids.*

Proof. Following a method from [7] let us begin with the notation of f being the total number of faces, e the total number of edges and v being the total number of vertices in a regular polyhedron. Let n denote the degree of a face, and d denote the degree at each vertex.

To have a polyhedron we require $n \geq 3$ and $d \geq 3$. As we are considering regularity n will be the same at all points in the polyhedron, as will d .

As each edge is adjacent to 2 faces, if we multiply f by n to find the total number of edges, we would have counted each edge twice, and so we have

$fn = 2e$. Similarly, as each edge is adjacent to 2 vertices if we multiplied v by the degree d we would count each edge twice, giving us $vd = 2e$.

Substituting the values for f and v that these two equations give us into Euler's formula leads to,

$$\frac{2e}{d} - e + \frac{2e}{n} = e\left(\frac{2}{d} - 1 + \frac{2}{n}\right) = 2.$$

As $e > 0$, $\frac{2}{d} - 1 + \frac{2}{n}$ must too be > 0 so that the answer 2 can be reached. This leads us to the inequality, $\frac{1}{d} + \frac{1}{n} > \frac{1}{2}$.

Let us first consider $\frac{1}{d} > \frac{1}{2} - \frac{1}{n}$. As we require $n \geq 3$ we have $\frac{1}{n} \leq \frac{1}{3}$. Substituting this into our formula we get,

$$\frac{1}{d} > \frac{1}{2} - \frac{1}{3} = \frac{1}{6}.$$

This leads us to $d < 6$. We can also apply exactly the same argument to $\frac{1}{n} > \frac{1}{2} - \frac{1}{d}$ to gain $n < 6$. As they are both ≥ 3 we have that $n = 3, 4$ or 5 and $d = 3, 4$ or 5 .

Now let us examine the nine possible combinations of n and d that this gives us, substituted into $\frac{1}{n} > \frac{1}{2} - \frac{1}{d}$:

n	d	implies		
3	3	$\Rightarrow$	$\frac{1}{3} > \frac{1}{6}$	✓
3	4	$\Rightarrow$	$\frac{1}{3} > \frac{1}{4}$	✓
3	5	$\Rightarrow$	$\frac{1}{4} > \frac{3}{10}$	✓
4	3	$\Rightarrow$	$\frac{1}{4} > \frac{1}{6}$	✓
4	4	$\Rightarrow$	$\frac{1}{4} > \frac{1}{4}$	×
4	5	$\Rightarrow$	$\frac{1}{5} > \frac{3}{10}$	×
5	3	$\Rightarrow$	$\frac{1}{5} > \frac{1}{6}$	✓
5	4	$\Rightarrow$	$\frac{1}{5} > \frac{1}{4}$	×
5	5	$\Rightarrow$	$\frac{1}{5} > \frac{3}{10}$	×

So we can see that there are only 5 valid combinations, and these represent the 5 platonic solids. In descending order from the properties in the table, they are called the tetrahedron, the octahedron, the isosahedron, the cube, and the dodecahedron.

□

Returning to consider all polyhedra, it was found by mathematician Lhuillier in 1811 that there are three exceptions where Euler's formula, as we know it, does not hold [2, p.85].

The first exception is when a polyhedron has an interior cavity which is completely enclosed. Imagine a hollow pyramid inside of a cube.

Theorem 27. *For a polyhedron that fully encloses any number of polyhedra inside of it, Euler's formula becomes $f + v - e = 2(n + 1)$ where n denotes the quantity of enclosed polyhedra.*

Proof. Let f , e and v denote the total number of faces, edges and vertices in the outer surface. Let f' , e' and v' denote the total number of faces, edges and vertices of the enclosed inner surface. Finally, let F , E and V denote the totals for the overall polyhedron.

Then we have $f + v - e = 2$ and $f' + v' - e' = 2$. Adding these together we gain $(f + f') + (v + v') - (e + e') = 4$. Then $F = f + f'$, $E = e + e'$ and $V = v + v'$, leaving us $F + V - E = 4$.

If we have an additional second enclosed cavity we can see, when following the same method, that Euler's formula will equal 6. Every new enclosed cavity adds 2 to the right hand side of Euler's formula in this way. Let n denote the number of cavities, then in general we find that $F + V - E = 2 + 2n = 2(n + 1)$. □

The next exception is when we have an indentation on the surface of a polyhedron formed from another one. Think a crater on the moon.

Theorem 28. *For a polyhedron that has any number of indented polyhedra on its surface, Euler's formula becomes $f + v - e = 2 + n$ where n denotes the quantity of indented polyhedra.*

Proof. Use the notation f , e and v as before for a polyhedron that contains an indent, let the dashed equivalents f' , e' and v' be associated to the polyhedron that forms the indent, and F , E and V denote the totals for the overall polyhedron.

Then we have $f + v - e = 2$ for the main polyhedron still, but we have $(f' - 1) + v' - e' = 1$ for the indent as it has one less face. Adding the two we get, $(f + f' - 1) + (v + v') - (e + e') = 3$. Then $F = f + f' - 1$, $E = e + e'$ and $V = v + v'$, leaving us $F + V - E = 3$.

We can see that each indent will increase the Euler formula by 1 and so we get that $F + V - E = 2 + n$.

□

The third and final case is probably the most useful one as it links to the torus, as we shall soon see. This exception arises when a tunnel passes through the polyhedron creating a hole. These tunnels are formed from polyhedra but are ‘missing’ their end faces to create an opening. Edges are also added to the faces from which the tunnel has its entrances, these edges are adjacent to the vertices of the tunnel and join to the vertices in the end face (See Figure 5.2 for an example). Note that without these extra edges Euler’s formula is satisfied as per normal, the presence of the edges is what creates the exception.

Theorem 29. *For a polyhedron that has any number of tunnels, with edges adjoining them to its exterior surface, Euler’s formula becomes $f + v - e = -2(n - 1)$ where n denotes the quantity of tunnels.*

Proof. Just like before, let f , e and v be assigned to the main polyhedron and this time f' , e' and v' will denote the total values for one component once we split it up.

Next imagine that we split our polyhedron in such a way that we are left with two new polyhedron that contain no tunnels. These are now not of any of the troublesome forms found by Lhuillier, and so will satisfy Euler’s formula as normal, $f' + v' - e' = 2$.

In doing this split each of the two new polyhedra have introduced 2 new faces. They have also gained d number of edges, where d is the sum of the edges in the new faces. Similarly we have d new vertices, where d is the sum of vertices in the new faces. Therefore, to fuse these back together we must remove the new additions.

Then we have that $f = 2(f' - 2)$, $e = 2(e' - d)$ and $v = 2(v' - d)$. As we have that $f' + v' - e' = 2$, we know that adding these two parts gives us

$2f' + 2v' - 2e' = 4$. Substituting e', v' and f' from the relations above gives us, $f + 4 + v + 2d - e - 2d = 4$. Once simplified we are left with, $f + v - e = 0$.

So we can see that the addition of a tunnel has decreased the Euler formula by two, leading us to the general form $f + v - e = 2 - 2n = -2(n - 1)$.

□

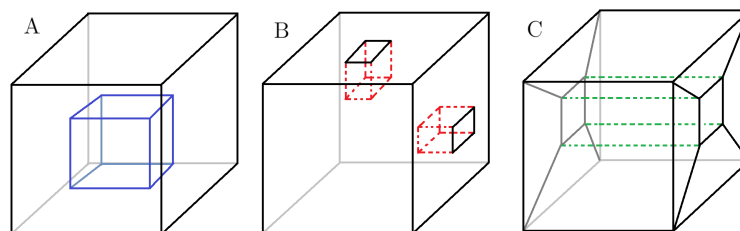


Figure 5.2: Visual examples of the three exceptions found by Lhuillier involving a cube.

Figure 5.2 contains cube A which has one cavity and has $v = 16, e = 24, f = 12$ giving us $16 + 12 - 24 = 4$ as expected by Theorem 27. Cube B has 2 indents and gives us $24 + 16 - 36 = 4$ as expected by Theorem 28. Cube C has one tunnel (note the extra edges) and gives us $16 + 16 - 32 = 0$ as expected by Theorem 29.

5.2 Surfaces

Returning to surfaces we will now consider the sphere and torus. A **smooth** surface is one that is infinitely differentiable [33, p.58]. We can just think of it as a surface without sharp edges and vertices, it is literally smooth.

A surface is **orientable** if it has a clear inside and outside. If you take a rectangle, twist it once and then stick the ends together a Möbius band is formed [3, p.276]. Tracing a line around it you would begin on the outside, enter the inside via the twist, and resurface on the outside, and so it is nonorientable. A sphere would be orientable.

A surface that is **closed** has no boundaries. Combining these three properties, smooth, orientable and closed gives us a group of surfaces that are defined by their **genus** [33, p.58]. A genus is the number of holes that they have, a sphere has genus 0, a doughnut has genus 1 and is more formally known as a torus, for example. These are denoted S_k where k is the genus number.

Let us begin with a discussion of S_0 surfaces. Through a process called **stereographic projection** we can see that planar graphs are embeddable on a sphere [3, p.247]. If we consider a sphere that is sat upon a flat plane P , then the point that is furthest away is where the projection stems from. The point that touches P can be thought of as the south pole, and the point furthest away as the north pole.

If this sphere had a graph embedded upon it we can stereographically project it onto P by drawing a line from the north pole, through a vertex v and onto the flat plane P . At the point that the line meets P the projection of v is drawn. This is repeated for all vertices, and the edges are maintained as per the spheres incidence function. It is evident to see that this results in a planar embedding of the spheres embedding.

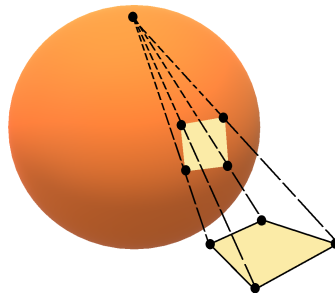


Figure 5.3: How stereographic projection stemming from the north pole works.

We can also see how we could apply the process backwards. If a sphere was sat upon a planar embedding then a line could be drawn up towards the north pole, this would hit the sphere at a point and the projection has found its desired location on the sphere. Therefore, we have that spherical embeddings can be drawn as planar embeddings, and planar embeddings can be drawn as spherical embeddings, and all will be 4-colourable.

We can also use this idea of projection to find planar embeddings of polyhedra [34, p.48]. If we engulf a polyhedra in a sphere and project from the north pole in the same way as before the result is a planar embedding, see Figure 5.4.

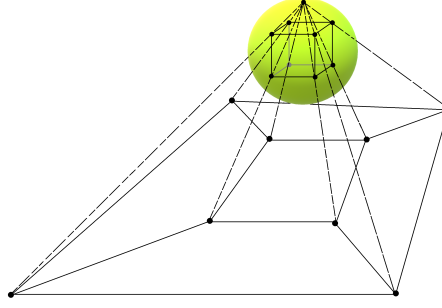


Figure 5.4: Projecting a cube.

It is this method that allows us to see why having the extra edges in the polyhedra case for Theorem 29 resulted in a form that would not satisfy the Euler formula when the formula is set to satisfy connected planar graphs. If we proceeded to draw the projection without the extra edges we would succeed in a planar embedding, however with the extra edges we would not be able to project it without edges crossing over each other. Therefore, it is not a planar graph and that is why a new form of Euler's formula was required.

Moving onto S_1 surfaces now, we have already seen in Theorem 29 that a tunnel in a 3-dimensional polyhedron resulted in $v + f - e = 0$ for Euler's formula. This tunnel can be thought of as a genus and it can be shown that the formula applies to all S_1 surfaces too [33, p.58].

Theorem 30. *A connected graph G that can be embedded onto a torus with simply-connected faces satisfies $f + v - e = 0$. Where f is the total number of faces, e is the total number of edges and v is the total number of vertices.*

Proof. Firstly, we need to note that each face must exist in such a way that it does not contain the hole of the torus, i.e. we do not need to cut it up in order to flatten it out on a plane. This type of face is called **simply-connected** [33, p.58]. An example of a face that isn't simply connected is one that wraps around the torus band joining back up with itself.

Proceeding with a proof by induction on the number of vertices and on the number of edges in G , assume $f + v - e = 0$ holds true for all v and e in G .

For a vertex with degree one, its edge will be incident to only one face. Then removal of this vertex and corresponding edge decreases e and v by 1 each and leaves f the same value, and so does not alter the formula.

For an edge that is incident to two faces, joining two vertices, its removal will decrease e and f by 1, again not affecting the formula. The new face formed by the removal of this edge will remain simply-connected as the two faces that it was formed from must have both been simply-connected too, so there will be an edge present to stop it looping fully around the hole.

It is evident then that the removal of these edges and vertices does not change the formula. We can repeat this until we are left with just one simply-connected face with all the remaining edges sitting on its boundary.

Opening up the torus along this boundary will leave us with a map that looks like Figure 5.5.

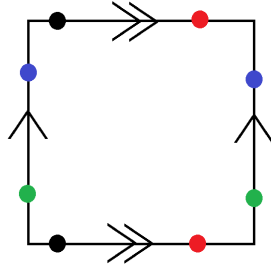


Figure 5.5: Ignoring the coloured vertices for now, this is a representation of a torus as by joining the edges labeled $>>$ together and then joining the edges labeled $>$ we build an S_1 surface. Then for our proof we have that all the edges are on the boundary of the face. The paired colours indicate which vertices are the same once the torus is formed.

We are left with m edges along the vertical line, and n edges along the horizontal. In total we have $m + n$ edges and $m + n - 1$ vertices and we only have one face. We have that there is one extra edge to vertices from the requirement of it being simply-connected. Collecting these into the formula gives us $m + n - 1 + 1 - m - n = 0$.

□

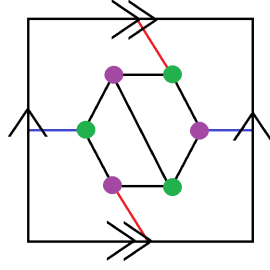


Figure 5.6: It is sometimes possible to draw nonplanar graphs onto other surfaces. Here we see the $K_{3,3}$ graph that we have previously proven is nonplanar, successfully embedded onto a torus. Checking Theorem 30 we have, $v + f - e = 6 + 3 - 9 = 0$.

As we have now seen that Euler's formula can have multiple results we shall rename it as **Euler's characteristic** and denote it as c for $v + f - e = c$ [3, p.279] (with v, f and e following our usual meaning). Then let a general closed surface be denoted by Σ with characteristic $c(\Sigma)$, and chromatic number $\chi(\Sigma) = k$ as defined previously. For example, when our surface is the sphere we have $c(\Sigma) = 2$ and $\chi(\Sigma) = 4$.

We will now make use of some previous theorems to find the chromatic number of a graph embedded upon a general closed surface Σ .

Recall from Theorem 13 that a k -critical graph G has $\delta(G) \geq k - 1$. We can think of our Σ as being k -critical as we are seeking the maximum value the chromatic number k needs to be. It is clear to see that the average degree in a graph G , denoted $d(G)$ (we will just use d for simplicity), will equal $d = \frac{2e}{v}$, and that the smallest degree $\delta(G)$ will be $\delta(G) \leq d$. Combining this with the result from Theorem 13 gives us $d + 1 \geq k$.

Next recall Lemma 1 where we found that a maximal planar graphs satisfies $v - e + \frac{2}{3}e = 2$. Using the idea of Euler's characteristic this becomes $v - e + \frac{2}{3}e = c$. Simplifying gives us, $e = 3(v - c)$ for maximally planar graphs or $e \leq 3(v - c)$ for planar graphs.

Substituting $e = \frac{dv}{2}$ from the average degree definition we get, $d \leq 6 - \frac{6c}{v}$. Then, we can combine this result with $d + 1 \geq k$ to get $k \leq 7 - \frac{6c}{v}$. Or equivalently,

$$\chi(\Sigma) \leq 7 - \frac{6c}{v}.$$

Thus for example, we can see that for a torus with $c = 0$, $\chi(S_1) \leq 7$.

In addition Heawood produced **Heawood's Inequality** that further solidified this result [3, p.392]. This inequality, for $c \leq 0$ is,

$$\chi(\Sigma) \leq \frac{1}{2}(7 + \sqrt{49 - 24c}).$$

This result comes from observing that $\chi(\Sigma) = k \leq v$, so that $k \leq 7 - \frac{6c}{v}$ becomes $k \leq 7 - \frac{6c}{k}$. Which when multiplied by k and rearranged gives us $k^2 - 7k + 6c \leq 0$. Solving this results in Heawood's Inequality.

Theorem 31. *Maps on a torus are 7-colourable.*

Proof. Heawood, in the same report from 1890 in which he disproved Kempe's four colour theorem 'proof', found a torus that had 7 faces that all neighboured one another [34, p.132]. Hence, 7 colours are required for this torus and when combined with our previous result was sufficient proof that tori are 7-colourable. $\square$

In addition to these ideas it is possible to find colouring theorems relating to nonplanar graphs. For example, we have the following Brooks' theorem (which we shall not show a proof for) [3, p.360].

Theorem 32. *(Brooks' Theorem) Any simply connected graph G is Δ -colourable, unless G is isomorphic to $K_{\Delta+1}$ or is an odd cycle.*

We have now seen how concepts of graph theory can be extended into other discussions beyond planar graphs, and this concludes our final objective.

Chapter 6

Conclusion

This project aimed to introduce the reader to graph theory and to then guide them into achieving an understanding of how to apply its ideas to prove the four colour theorem. In Chapter 2 we saw that this is not such a simple task. The history behind the theorem revealed that finding the proof was a very long process, taking many mathematicians many years to reach a widely undisputed proof, made possible by the use of computers. Even now there are still a handful of ‘traditional’ mathematicians who believe a proof is only valid if seen on paper and calculated by hand [33, p.135].

We hope that our introduction to graph theory in Chapter 3 was digestible and enjoyable. We visited other well known theorems and problems such as the Königsberg problem that birthed graph theory, the three utilities problem and Hall’s marriage theorem. These discussions helped to build up enough concept knowledge to tackle our main discussion in Chapter 4.

The main result we wanted to reach in this chapter was an understanding of reducible unavoidable sets found through a method of discharging. These ideas were the crux of Appel & Hakens proof of the theorem in 1976. Before we discussed these we also uncovered proofs for Kuratowski’s theorem, the six colour theorem and the five colour theorem. All of the ideas in this chapter combined to reveal how the theorem was proved and allowed a deeper understanding of why it took so long for this to be achieved.

To conclude, we began considering how graph theory and colouring theorems can be used in other forms. Beginning with polyhedra and moving onto surfaces we found that many concepts were easily transferable. We saw that a torus is seven colourable, and surprisingly to us, this fact was significantly easier to find than discovering that a plane is four colourable.

To further this project as it is we would continue the Chapter 5 topics, looking into more surface types and theorems. Additionally, we want to further explore any proofs for theorems or ideas that were omitted in the project, in particular the Jordan curve theorem.

To expand the study we would like to investigate the real world applications of graph theory. We briefly noted in the introduction that it has its uses and we have found that the most commonly referenced one is for networking considerations. We were fortunate enough to attend a lecture during the time that we were researching this project. This showed how brain signals can be mapped as if they were a bipartite graph (combined with linear algebra concepts) with one set containing the neurons that send a signal and the other set being the muscles that receives it [8]. Also, we have come across ideas that use the four colour theorem itself. For example, [9] discusses if the four colour theorem can be used to see how ecosystems should be best arranged and if this in turn could help with conservation biology.

Overall we hope that our project aims and objectives have been realised and that you, the reader, have grasped an understanding of the four colour theorem and the graph theory that surrounds it. Thank you for reading.

Chapter 7

Acknowledgements

I extend my warmest gratitude towards my supervisor [REDACTED] [REDACTED] [REDACTED]. Your support and guidance, not only during this project but throughout my degree, has allowed me to enjoy researching and writing this project and to produce something that I am proud of. Thank you.

I also thank all of the lecturers in the University of Lincoln School of Mathematics and Physics department that have taught me since day one of my degree. Without your knowledge, encouragement and patience I could not have even begun this project.

Lastly I would like to acknowledge my friends and family. Thank you all for your support and love, it has kept me motivated when I would otherwise be distracted. In particular, I thank [REDACTED] who have all endured my many graph theory orientated conversations and proof reading requests. I appreciate all that you have done for me.

Chapter 8

Appendices

This chapter provide copies of the research plan and midway progress report, followed by a table summarising the notation used throughout the report.

8.1 Research plan

3rd Year Mathematics Project Research Plan by

██████████

The aim of this research plan is to outline the intended approach to undertaking the 3rd year mathematics project. The project outline will be discussed, why the project was chosen and what research and literature will be considered. A brief action plan with a timeline will also be determined. This is to help guide and keep the project on track throughout the entirety of the course.

8.1.1 The project

The chosen project is titled **The 4-colour theorem** and ██████████

██████████ This is the theorem that given a map, you need no more than 4 colours to be able to colour all regions so that no two adjacent regions have the same colour, see Figure 8.1 for an example. It is a theorem that puzzled for centuries and was only finally solved with the help of a computer.

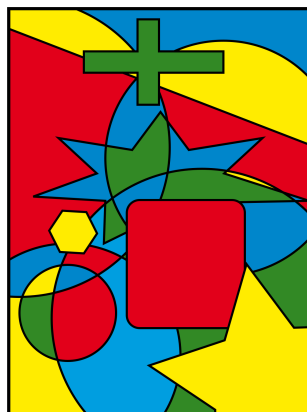


Figure 8.1: A four coloured map [31].

The 4-colour theorem revolves around **graph theory**, and so that is where this project will begin. The main plan is to cover a background knowledge of graph theory, including examples and proofs for theorems that are come across. This will be built up until the topics that link to the 4-colour theorem have been discovered, ready for that to then be discussed. Other concepts of further interest will also be discussed as they are learned and unravelled.

This project was chosen as it closely links to a topic from the 2nd year Group Project module. In that project, the aim similarly was to build up a background knowledge for graph theory, with the final target of proving Hall's Marriage Theorem. It was a really enjoyable project as it was a completely brand new branch of mathematics being explored by the group at the time. It is hoped to take this previous experience with it and apply it in a more complex setting. Some concepts will undoubtedly be revisited but will be explored further and expanded upon.

8.1.2 Action plan

Now the action plan for the project will be set out. The aim is to stick to this schedule, however, it is subject to change as it may become apparent that it is beneficial to consider a different structure as the project develops.

Week number	Work Target
1-3	• Background reading of literature • Completion of project plan
4-7	• Introduction to graph theory, including the history and attempts at solving the 4-colour theorem • Begin writing of background knowledge • Types of graphs, including examples and any theorems given with their proofs
7	• First logbook due
8-11	• Planar graphs (including Euler's formula) and chromatic numbers to receive a more in depth section each • Discussion of the colouring theorems, including 6 and 5-colourable proofs
12	• Second logbook due • Progress report due week 13, to be written this week

13-16	• Semester 2 revision and Christmas break • Project put to aside mainly, background knowledge section expanded upon
17-26	• Completion of semester 1 targets • Research into graphs on other surfaces • Any further topics – applications etc. • Abstract, introduction and conclusion • Acknowledgements
26	• Draft report submitted
26-30	• Proof read, proof read, proof read! • Check references match • Add research plan/progress report as appendices • Viva preparation
31	• Final report due
32	• Viva presentation due

In addition to this plan, the arrangements for the supervisor meetings are to meet at 1pm Thursdays during the 1st semester, with the exception of week 9 which will be held Tuesday 1pm. Going forward, meetings are provisionally booked in for 1pm Tuesday during the 2nd semester.

8.1.3 Survey of the literature

To conclude, the literature already in use will now be surveyed. This survey will briefly state what was enjoyed about each piece and how and why it connects to the project. It is not expected that this will be the full list of all the literature used overall, but it does give a good base to start from.

J. A. Bondy and U.S.R Murty, Graph Theory, Springer, 2010.

The Bondy and Murty book is a comprehensive guide to Graph Theory. It covers all the background content that will be studied and provides clear examples and diagrams. With a dedicated chapter to the 4-colour theorem, it is expected to be the most useful resource. Subsection 10.6 covers surface embeddings of graphs, a topic that will be looked at further in weeks 17-26.

J. A. Bondy and U.S.R Murty, Graph Theory with Applications, North-Holland, 1982.

By the same authors as above but published before the 4-colour theorem was fully realised this book is not as helpful as the first. However it takes the basics to a more in-depth level and gives examples of applications, so those will be looked at to help investigate areas of interest further.

Jeremy L. Martin, The Notorious Four-Color Problem, University of Kansas, 2013.

This PowerPoint gives a really clear breakdown of the theorem with some useful examples. More importantly, it has a timeline of the significant historical moments regarding the 4-colour theorem so that will be used to further research later on.

Yuriy Brun, The Four Color Theorem, Undergraduate Journal of Mathematics.

This article again has some useful examples. It contains few theorems but they are explained clearly and the colourable ones will be useful. It also has a short reference list which will be investigated at a later date.

Georges Gonthier, Formal Proof – The Four-Color Theorem, Notices of the AMS, 2008.

This article includes some history of the theorem and discusses the backlash mathematicians received because they had used a computer to solve the theorem initially. It then discusses how to consider solving it without a computer and highlights the new concepts and theorems discovered by attempting to validate the computers proof. The idea of discussing the new theorems discovered will again be investigated at a later date.

8.3 Figures and notations

Unless referenced, all images were created for this project. Those using Europe were created using:

Blank Digital Map Of Europe, *Showyou*. [Online] Available: <http://showyou.me/blank-map-europe.html/blank-digital-map-of-europe-youreurope-map-com-best-europe>

For clarity and reference, the table below summarises the notation we have used throughout the report, along with its meaning.

Notation	Definitions
$G = (v, e)$	Graph G with v vertices and e edges
$v(G)$	Set of vertices in a graph G
$e(G)$	Set of edges in a graph G
$f(G)$	Set of faces in a graph G
$deg(v)$	Degree of a vertex v
$\Delta(G)$	Maximum degree of vertices in a graph G
$\delta(G)$	Minimum degree of vertices in a graph G
$N(v)$	Neighbourhood of a vertex v
ψ_G	Incidence function of a graph G
$G \cong H$	Graph G is isomorphic to graph H
$Aut(G)$	Set of automorphisms of a graph G
$aut(G)$	Quantity of automorphisms of a graph G
K_n	Complete graph with n vertices
N_n	Empty graph with n vertices
C_n	Cyclic graph with n vertices
W_n	Wheel graph with n vertices
S_n	Star graph with n vertices
$G[X, Y]$	Bipartite graph G with parts X and Y
$K_{m,n}$	Complete bipartite graph with subsets of size m and n
$G \subseteq H$	Graph G is a subgraph of graph H
$G \supseteq H$	Graph G is a supergraph of graph H
$G - v$	Graph G with a vertex v removed
$G \setminus e$	Graph G with edge e removed

$G \cup H$	Union of graphs G and H
$G \cap H$	Intersection of graphs G and H
$G \oplus H$	Ring of graphs G and H
$G + H$	Union of disjoint graphs G and H
$\bar{G}$	Compliment of a graph G
$\kappa(G)$	Vertex connectivity of a graph G
$\lambda(G)$	Edge connectivity of a graph G
$\tilde{G}$	Planar embedding of a graph G
G^*	Dual of a graph G
$\chi(G)$	Chromatic number of a graph G
$\mathcal{U}$	A set of configurations
S_k	Closed surface S with genus k
$x \in X$	x is an element of X
$ x $	Modulus of x

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